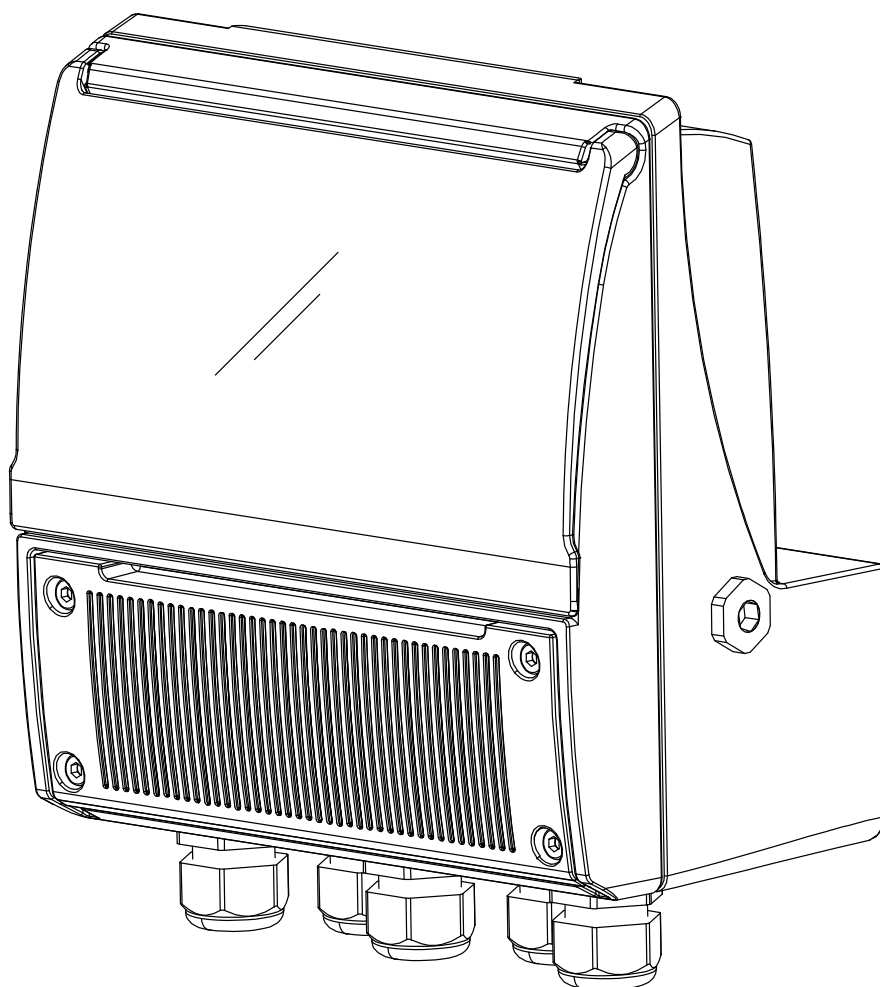




ISOMAG®

The friendly magmeter

OPERATING AND MAINTENANCE MANUAL



MV210

CE

ISOIL®
INDUSTRIA



Release number: **210_EN_IT_IS_R1_1.02.0**

The characters of file name in bold type indicate the software version which the manual refers to; it is visualized at the instrument start up, or by specific function on DIAGNOSTIC menu.



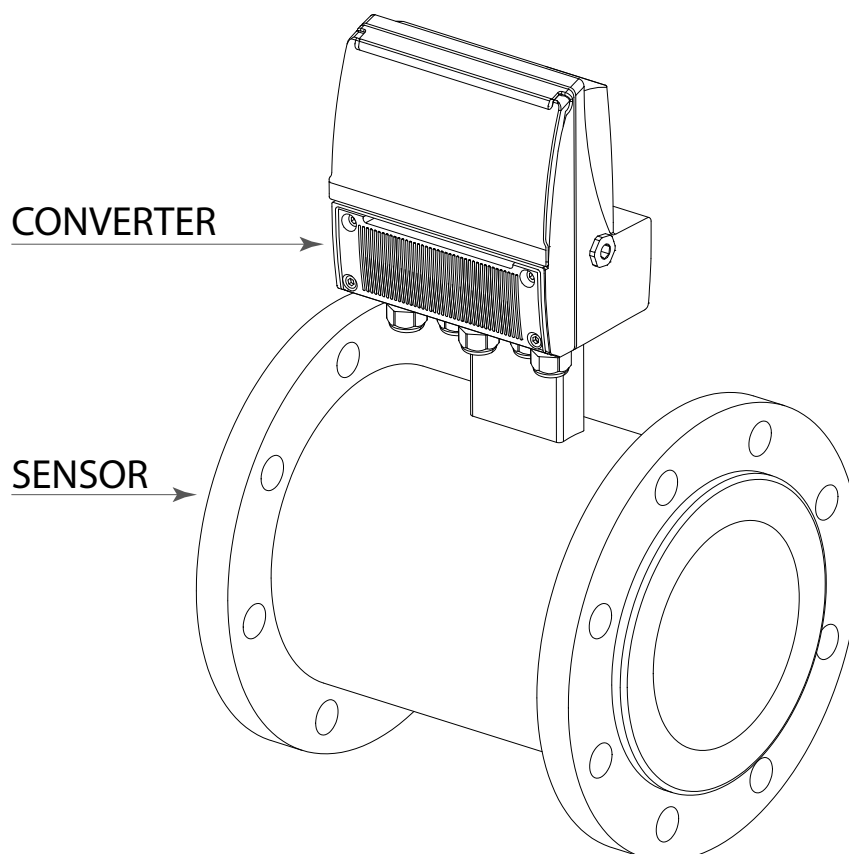
INDICE

MV210	1
SAFETY INFORMATION	3
INTRODUCTION	3
SAFETY CONVENTION	4
DATA PLATE	5
AMBIENT TEMPERATURE	5
TECHNICAL CHARACTERISTICS	5
DIMENSIONS	6
SEPARATE VERSION	7
HORIZONTAL VERSION	8
VERTICAL VERSION	8
MV210 CONSTRUCTION	10
INTERNAL LAYOUT	11
ELECTRICAL CONNECTION AND GROUNDING INSTRUCTIONS	12
INTERNAL LITHIUM BATTERY	13
DIGITAL INPUT ON/OFF OPERATION	15
OUTPUTS WIRING	16
DIGITAL OUTPUTS	16
ANALOG OUTPUTS	16
RS485 MODBUS MODULE (OPTIONAL)	17
CONNECTORS MIL	18
MEANING OF FLAGS	21
MEANING OF LED COLORS	21
ACCESS VIA KEYPAD	22
ACCESS VIA MCP INTERFACE (VIRTUAL DISPLAY)	22
ACCESS TO THE CONFIGURATION MENU	22
FLOW RATE VISUALIZATION	23
FLOW RATE ALERT	23
QUICK START MENU	23
ACCESS CODE SET : MENU 13 SYSTEM	24
RESTRICTED ACCESS SET : MENU 13 SYSTEM	24

CONVERTER ACCESS CODE	24
FUNCTIONS MENU	27
FUNCTIONS DESCRIPTION	32
MENU 1 - SENSOR	33
MENU 2 - UNITS	36
MENU 3 - SCALE	38
MENU 4 - MEASURE	40
MENU 5 - ALARMS	42
MENU 6 - INPUTS	43
MENU 7 - OUTPUTS	44
BATCHING	45
MENU 8 - COMMUNICATION	50
MV210 MODBUS PROTOCOL	50
MENU 9 - DISPLAY	53
MENU 10 - DATA LOGGER	54
MENU 11 - FUNCTION	61
MENU 12 - DIAGNOSTIC	62
MENU 13 - SYSTEM	65
MENU 14 - FILE (ONLY MCP)	71
MENU 15 - PROCESS DATA (ONLY MCP)	73
METER DATA	75
B.I.V. (BUILT-IN VERIFICATOR)	76
SOFTWARE ISOBIV	79
ALARM MESSAGES (CAUSES AND ACTIONS TO BE TAKEN)	80
ERROR CODE TEST SYSTEM OF SENSOR	81
MANUAL REVIEWS	83

INTRODUCTION

- ❑ These operating instructions and description of device functions are provided as part of the scope of supply.
- ❑ They could be modified without prior notice. The improper use, possible tampering of the instrument or parts of it and substitutions of any components not original, renders the warranty automatically void.
- ❑ The flow meter realizes a measure with liquids of conductivity greater than $5\mu\text{S}/\text{cm}$ in closed conduits, and is composed of a converter (described in this manual) and a sensor (refer to the specific manual).
- ❑ The converter could be coupled directly on the sensor (compact version) or coupled to the sensor by cable supplied with it (remote version).



SAFETY INFORMATION

Any use other than described in this manual affects the protection provided by the manufacturer and compromises the safety of people and the entire measuring system and is, therefore, not permitted. The manufacturer is not liable for damaged caused by improper or non-designated use.

- ❑ Transport the measuring device to the measuring point in the original packaging. Do not remove covers or caps until immediately before installation. In case of cartons packaging it is possible to place one above the other but no more than three cartons. In case of wooden packaging do not place one above the other.
- ❑ Disposal of this product or parts of it must be carried out according to the local public or private waste collection service regulations.
- ❑ The converter must only be installed, connected and maintained by qualified and authorized specialists (e.g. electrical technicians) in full compliance with the instructions in this Operating Instruction, the applicable norms, legal regulations and certificates (depending on the application).

- ☐ The specialists must have read and understood these Operating Instructions and must follow the instructions it contains. The Operating Instructions provide detailed information about the converter. If you are unclear on anything in these Operating Instructions, you must call the ISOIL service department.
- ☐ The converter should only be installed after have verified technical data provided in these operating instructions and on the data plate.
- ☐ Specialists must take care during installation and use personal protective equipment as provided by any related security plan or risk assessment.
- ☐ Never mount or wire the converter while it is connected to the power supply and avoid any liquid contact with the instrument's internal components. To connect remove the terminals from the terminal block.
- ☐ Before connecting the power supply check the functionality of the safety equipment.
- ☐ Repairs may only be performed if a genuine spare parts kit is available and this repair work is expressly permitted.
- ☐ For the cleaning of the device use only a damp cloth, and for the maintenance/repairs contact the service center (for details see the last page).

Before starting up the equipment please verify the following:

- ☐ Power supply voltage must correspond to that specified on the data plate
- ☐ Electric connections must be completed as described
- ☐ Ground (earth) connections must be completed as specified

Verify periodically (every 3-4 months):

- ☐ The power supply cables integrity, wiring and other connected electrical parts
- ☐ The converter housing integrity
- ☐ The suitable tightness of the sealing elements
- ☐ The front panel integrity (display and keyboard)
- ☐ The mechanical fixing of the converter to the pipe or wall stand

SAFETY CONVENTION

 DANGER ELECTRIC SHOCK	 WARNING	 PRECAUTIONS	 ATTENTION
--	--	---	--

TECHNICAL CHARACTERISTICS

ELECTRICAL CHARACTERISTIC



Converter classification: class I, IP67/68 for aluminum and PA6 housing, installation category (overvoltage) II, rated pollution degree 2.

Power supply versions	Power supply voltage	Power supply frequency	Min Power	Max power
HV	100-240V~	45-66HZ	1,5 W (Sensor only)	5 W (all Loads)
LV	24-36V---	//		
	24-36V~	45-66HZ		
LLV	12-48V---	//		

- ☐ Voltage variations must not exceed $\pm 10\%$ of the nominal one.
- ☐ Input/output insulated up to 500V.
- ☐ The output 4-20mA (optional) is electrically connected to the ON/OFF outputs and the output power supply (24V ---).
- ☐ Version LV/LLV : inrush current < 20A
Version HV : inrush current < 25A

ENVIRONMENTAL USE CONDITIONS



- ☐ The converter can be installed internally or externally
- ☐ Altitude: from -200m to 4000m (from -656 to 6560 feet)
- ☐ Humidity range: 0-100%

AMBIENT TEMPERATURE				
	Aluminium		Reinf. Nylon	
	Min*	Max	Min*	Max
°C	-20	+ 60	-10	+ 50
°F	-4	+140	+14	+122



ATTENTION

The battery will not be charged outside the below limits :

- ☐ T board MV210 < 0 °C
- ☐ T board MV210 > 50 °C

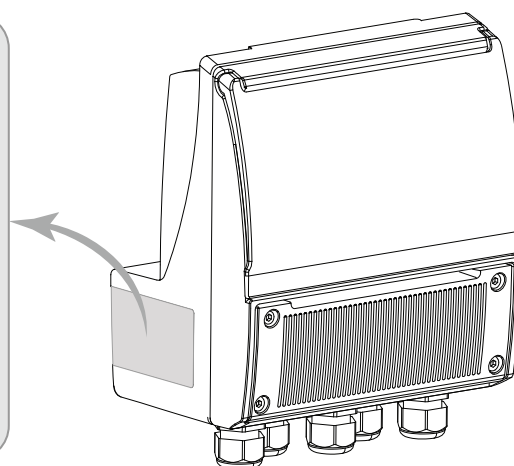
If the converter is supplied in compact version (converter over the sensor), consider the ambient temperatures more restrictive, otherwise refer to the relevant manuals.

* For discontinuous use, a thermostat heat source installation may be necessary.

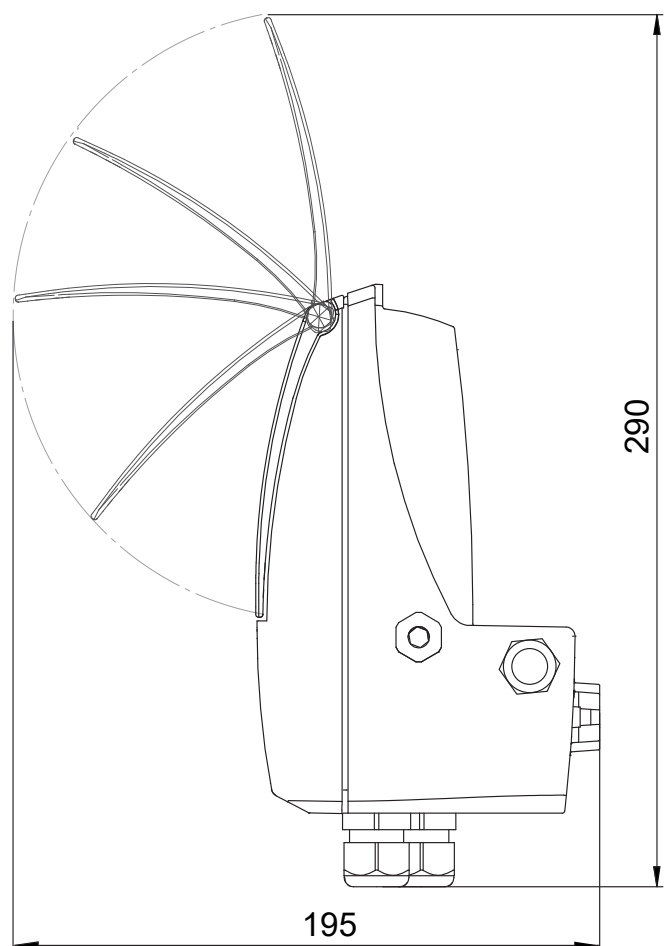
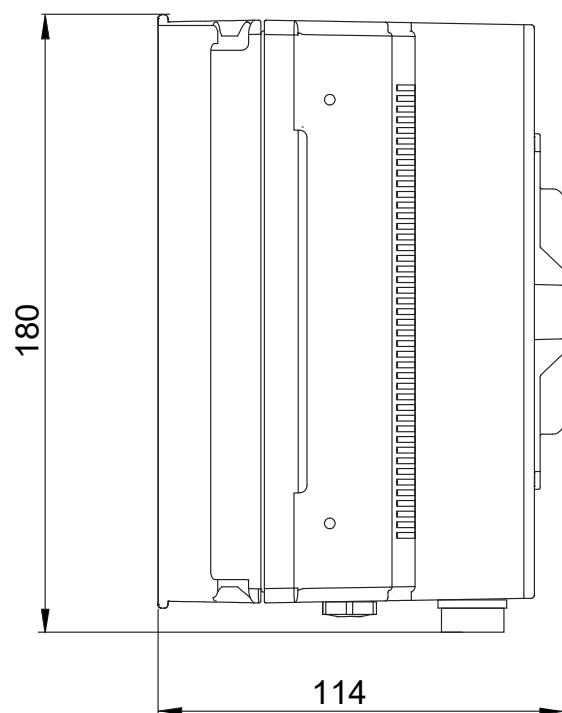
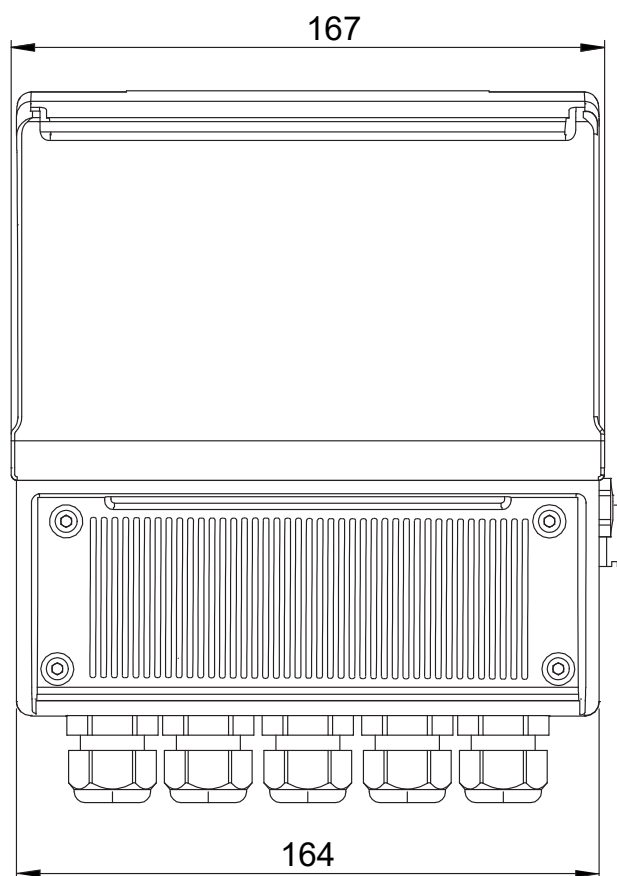
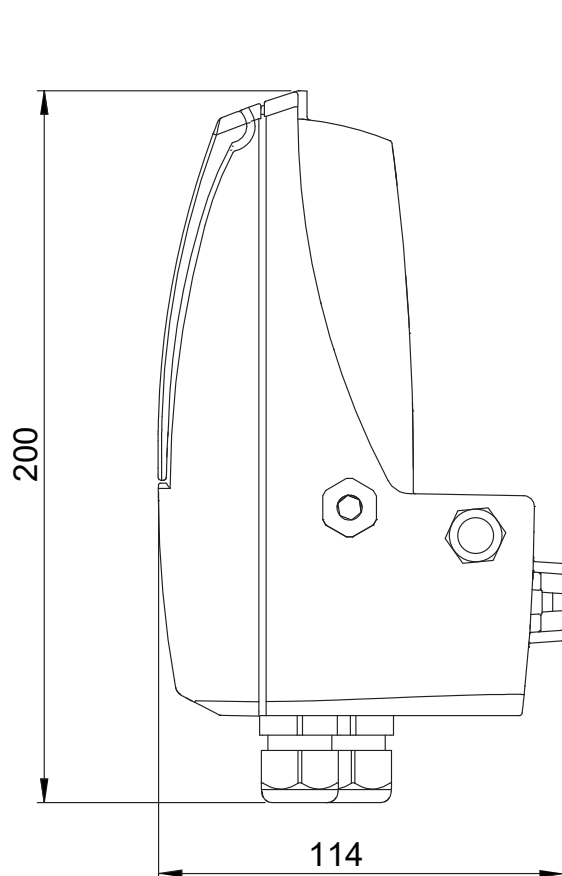
DATA PLATE

The instrument label contain the following information:

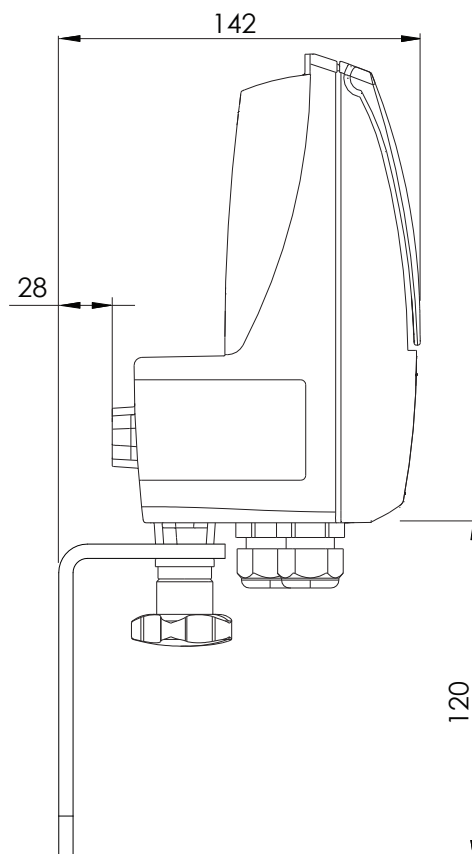
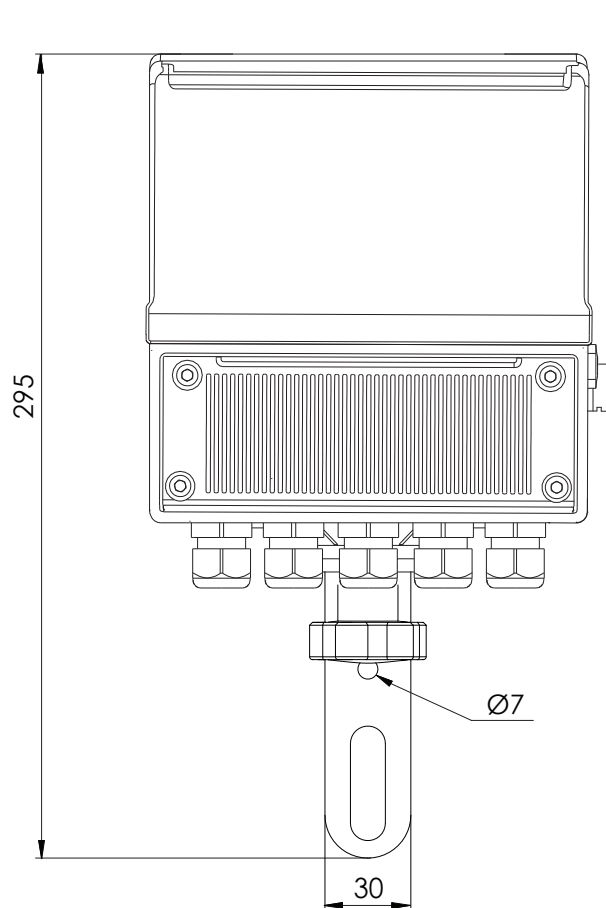
- ☐ MODEL: Convert Model
- ☐ S/N: Serial Number of the converter
- ☐ SUPPLY: Main power supply
- ☐ Hz: Supply frequency (AC)
- ☐ POWER: Maximum power consumption
- ☐ IP: Protection grade
- ☐ T: Operation temperature
- ☐ COUPLING: Serial number of sensor coupled
- ☐ ITEM: Free for user



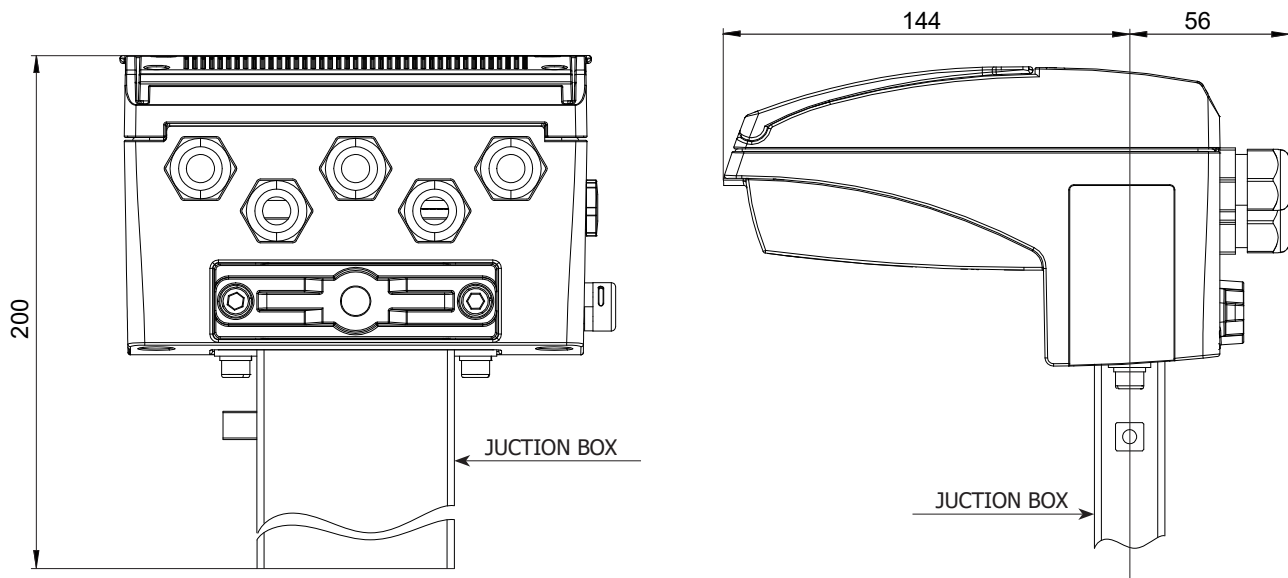
DIMENSIONS



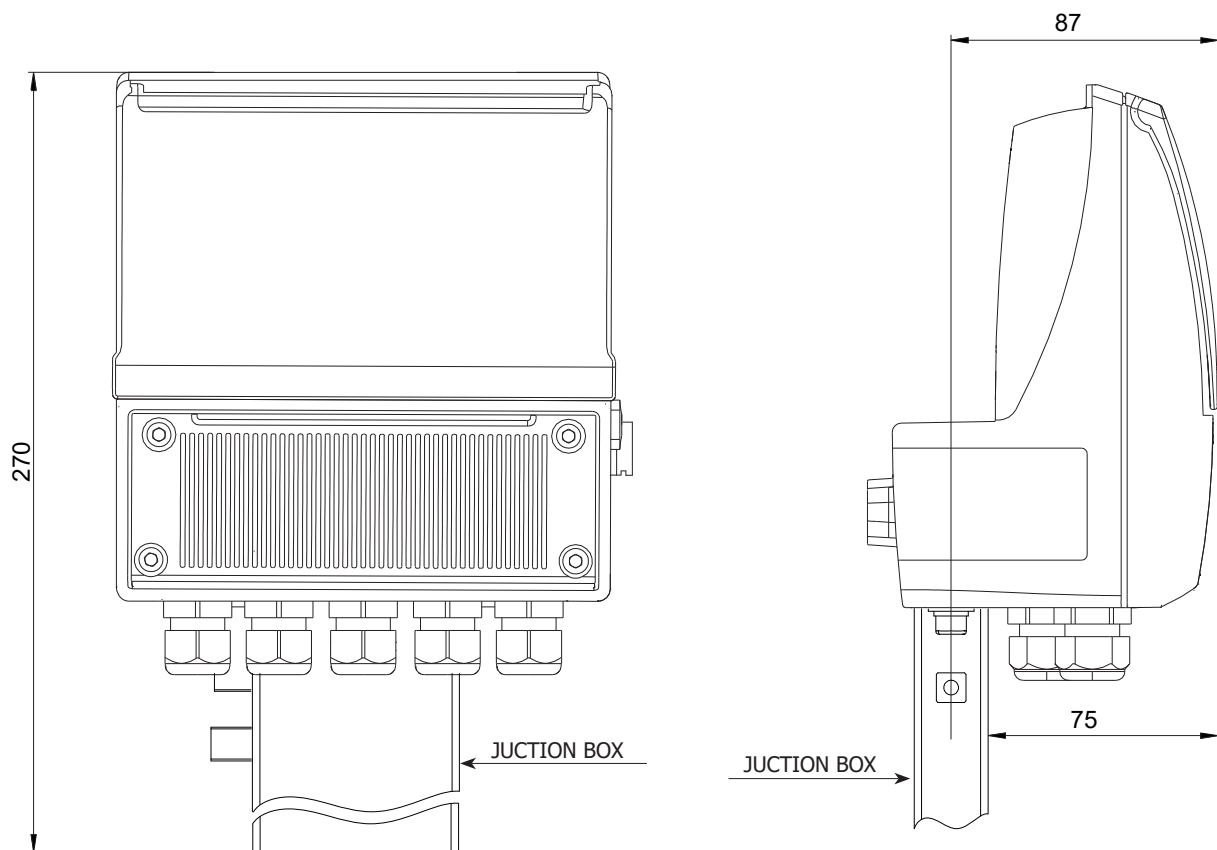
SEPARATE VERSION



HORIZONTAL VERSION



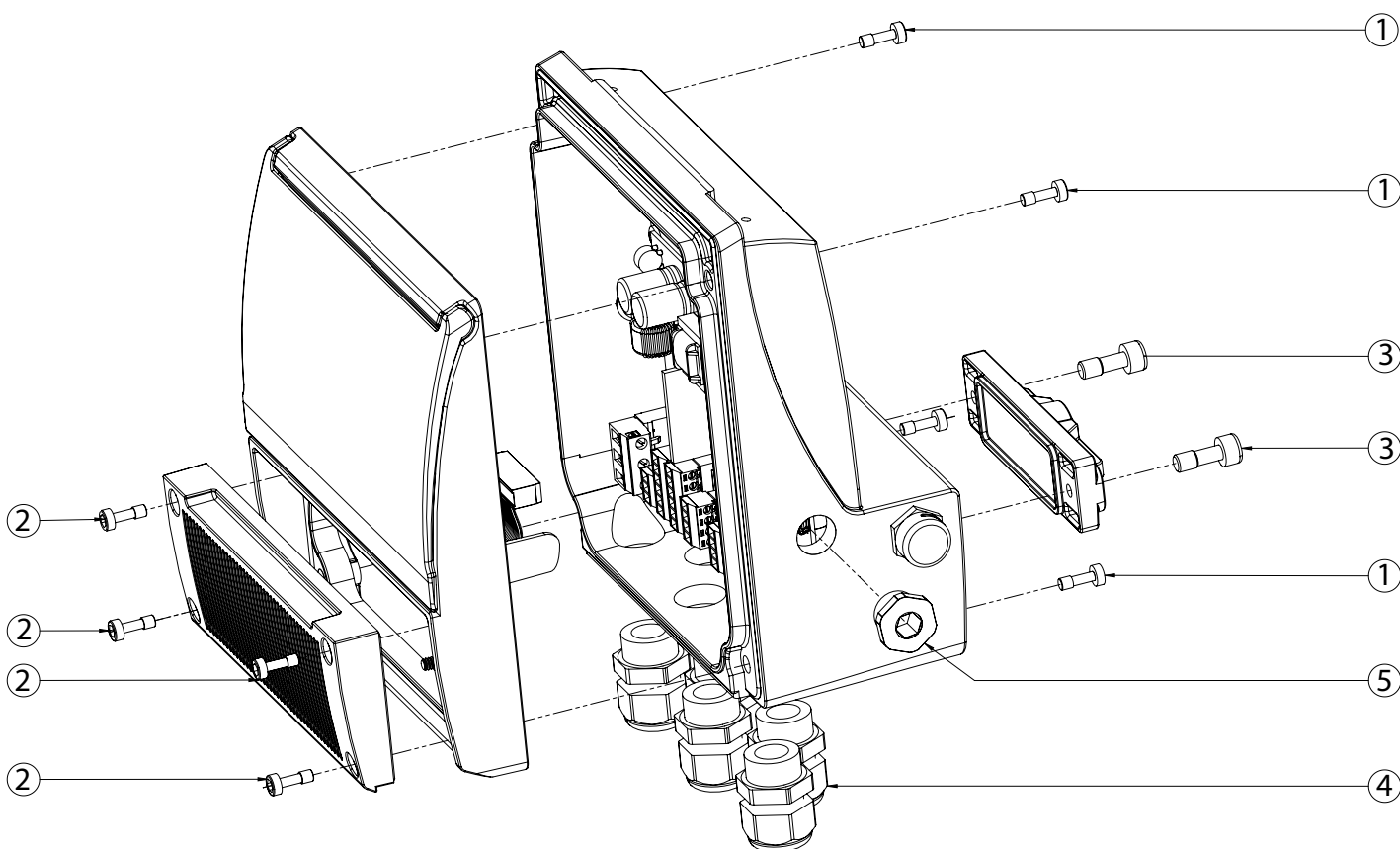
VERTICAL VERSION



TORQUES

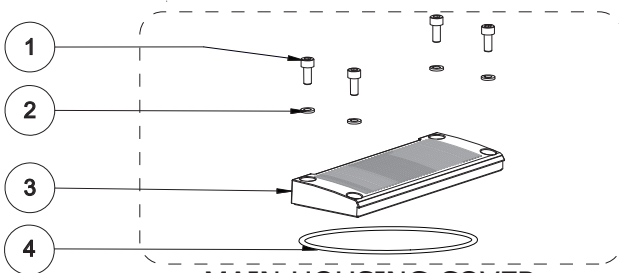
To guarantee the housing's IP degree the following torques are required:

HOUSING MATERIAL	Housing screws (1)	Screws Terminal block cover (2)	Fixing Display Frame	PCB Screws	Version Cap (3)	Cable Glands (4)	Cap USB-B (5)
ALUMINIUM HOUSING	6 Nm	5.5 Nm	3 Nm	0.8 Nm	8 Nm	4 Nm	4 Nm
PLASTIC HOUSING	2 Nm	2 Nm	2.5 Nm	0.8 Nm	7 Nm	4 Nm	4 Nm

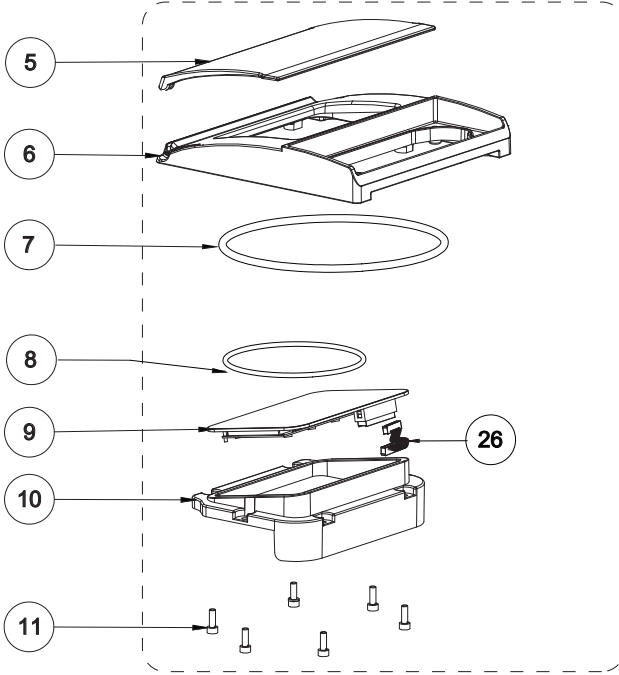


MV210 CONSTRUCTION

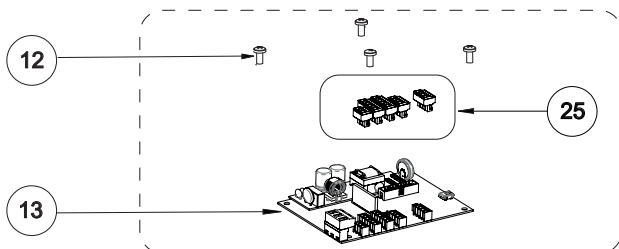
TERMINAL BLOCK COVER



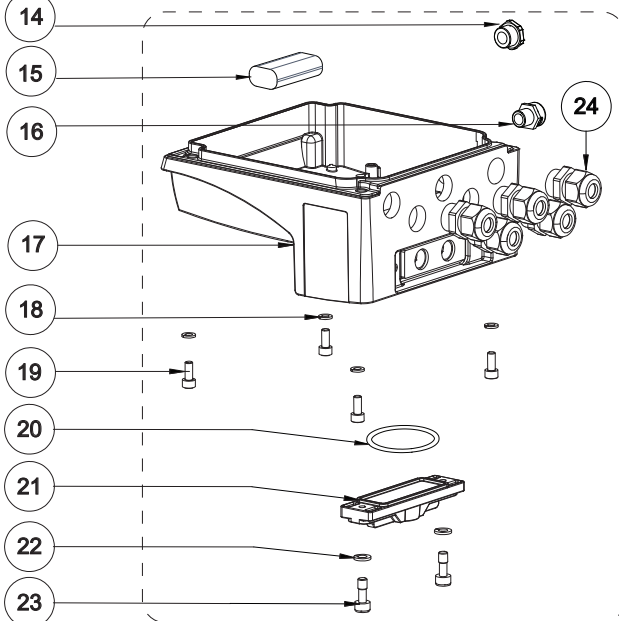
MAIN HOUSING COVER



PCB MV210



MAIN HOUSING

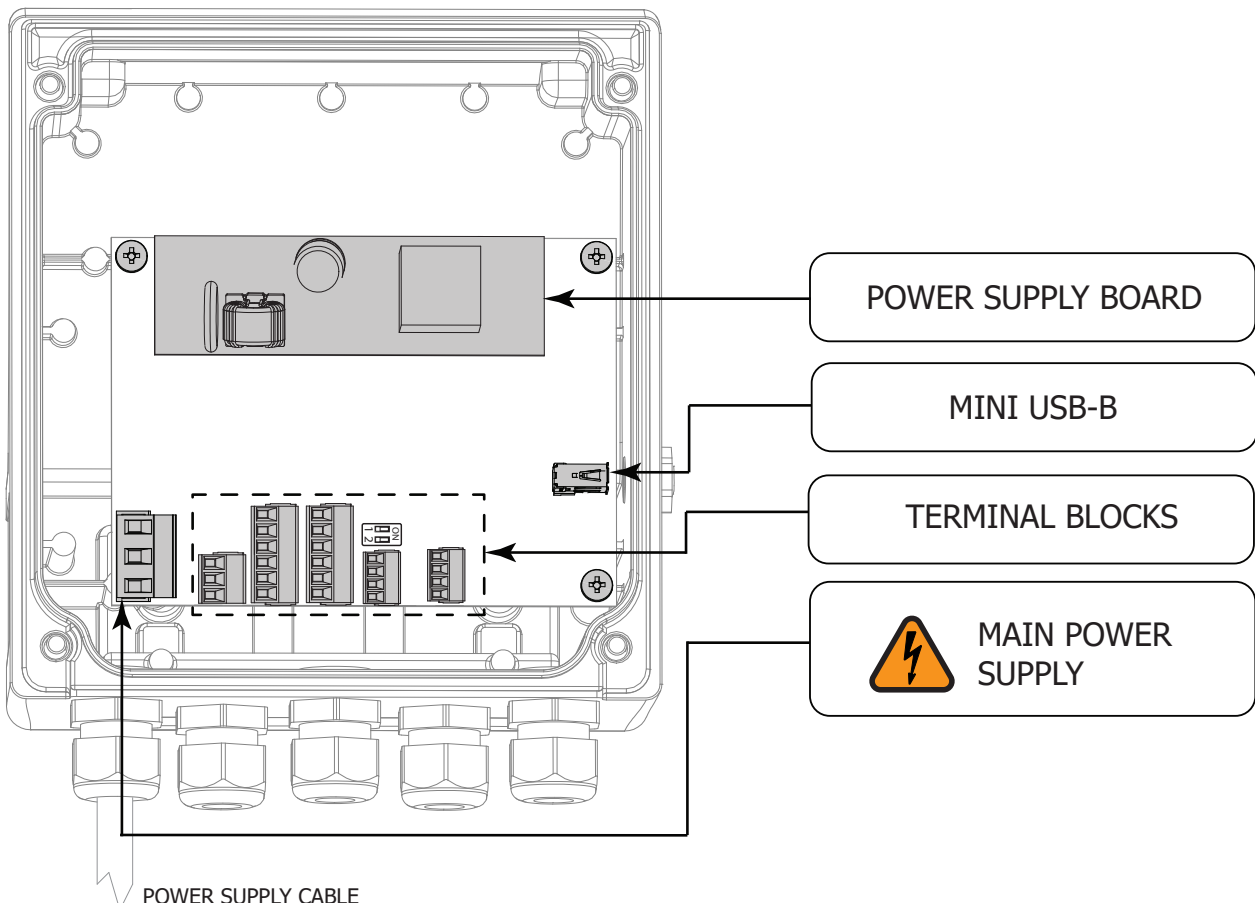
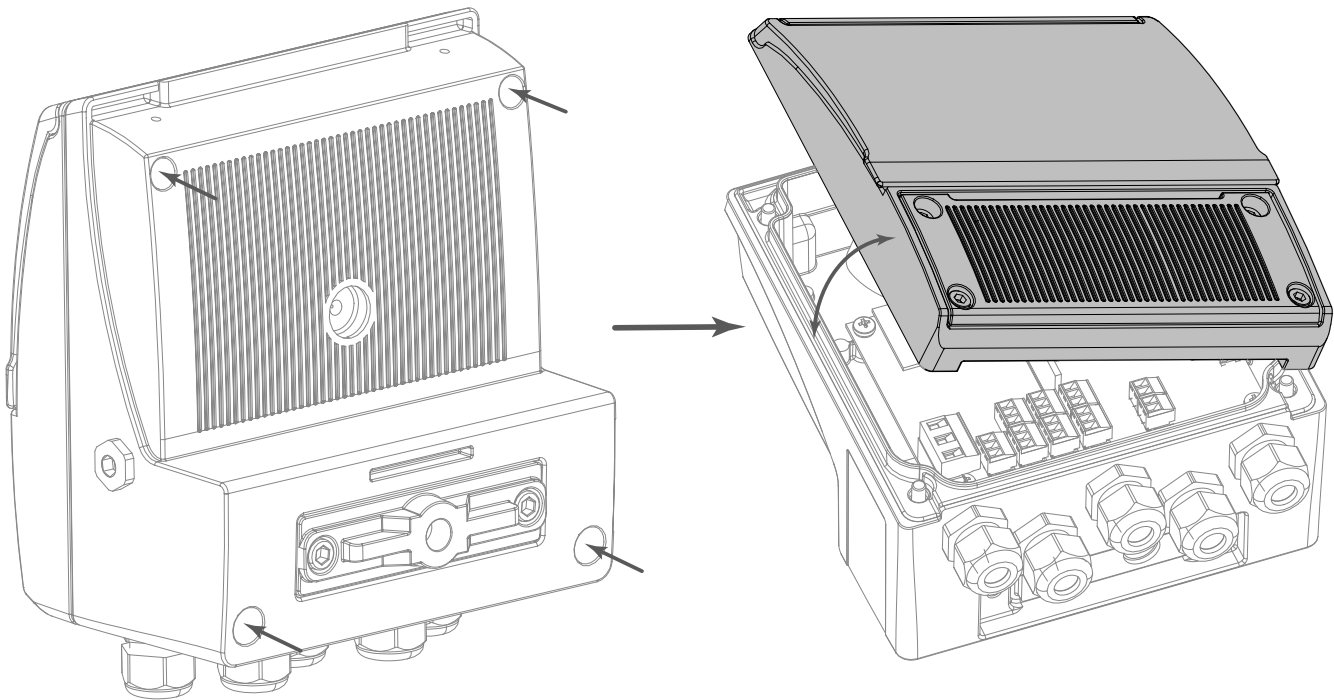


POS.	DESCRIPTION	
	PA6 VERSION	ALUMINIUM VERSION
1	SCREW M4x12	SCREW M5x12
2	GROWER Ø4	GROWER Ø5
3	TERMINAL BLOCK COVER	TERMINAL BLOCK COVER
4	O-RING-4400	
5	PROTECTION COVER	
6	HOUSING COVER	HOUSING COVER
7	ORING-4700	
8	ORING-117x3	
9	DISPLAY	
10	FIXING DISPLAY FRAME (MATERIAL PA06)	
11	SELF-TAPPING SCREW 4x10	TRILOBO SCREW 4x10
12	SELF-TAPPING SCREW 4x10	TRILOBO SCREW 4x10
13	PCB MV210	
14	PG9 CAP	
15	LITHIUM BATTERY	
16	ANTICONSENSE CAP	
17	PA6 MAIN HOUSING	ALUMINIUM MAIN HOUSING
18	GROWER Ø4	GROWER Ø5
19	SCREW M4x12	SCREW M5x12
20	O-RING-155	
21	VERSION CAP (MATERIAL PA06)	
22	SCREW M6x16	
23	GROWER Ø6	
24	PG11 CABLE GLAND CABLE DIAMETER: Ø5-Ø10mm	
25	TERMINAL BLOCK SOLID WIRE: 26-16 AWG / 0.129-1.31 mm² STRANDED WIRE: 26-16 AWG / 0.129-1.31 mm² TORQUE: 3.0 Lb.In / 0.34 Nm	
26	FLAT CABLE	

INTERNAL LAYOUT

INTERNAL CONVERTER VIEWS

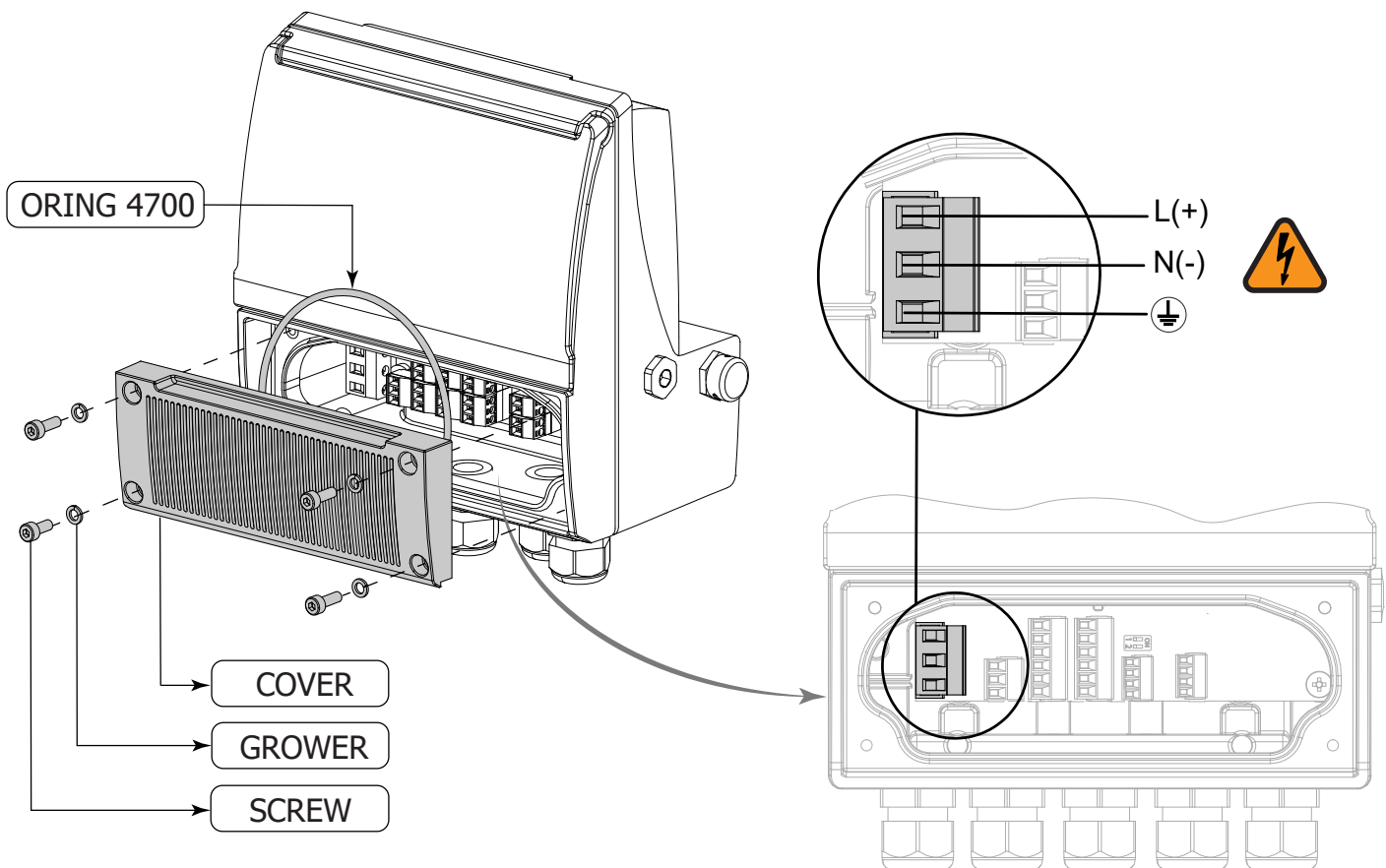
Remove the main housing cover by removing the 4 screws as shown here below.



ELECTRICAL CONNECTION AND GROUNDING INSTRUCTIONS



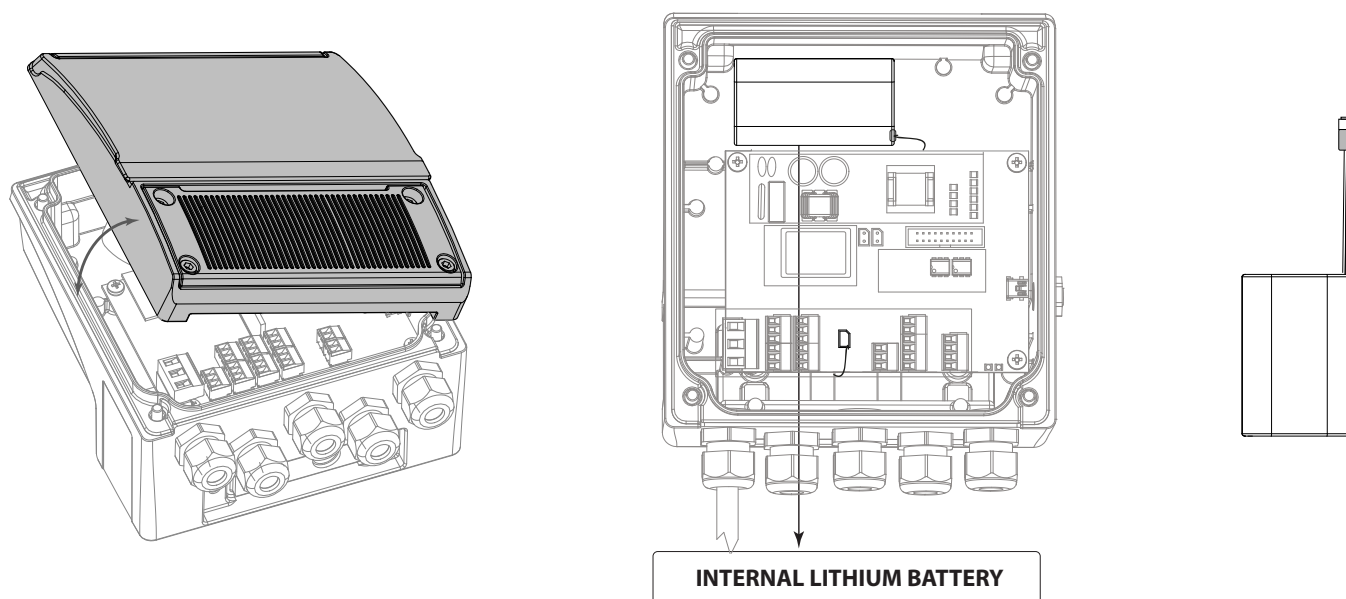
Always ensure that the converter and the sensor are grounded (earthed) correctly. The grounding of the sensor and converter **must** ensure that the instrument and liquid are equipotential.



- ❑ Before connecting the power supply, verify that the mains voltage is within the limits indicated on data plate.
- ❑ For the connections use only approved conductors, with fire-proof properties, whose section varies from 0.25mm² to 1.50mm², based on distance/power; additionally fix the power supply wires with a additional fastening system located close to the terminal.
- ❑ The power supply line must be equipped with an external protection for overload current (fuse or automatic line breaker).
- ❑ Provide in close proximity the converter a circuit breaker easily accessible for the operator and clearly identified; whose symbols must conform to the electrical safety and local electrical requirements.
- ❑ Ensure that the component complies with the requirements of the standard for electrical safety distance.
- ❑ Check chemical compatibility of materials used in the connection security systems in order to minimize electrochemical corrosion. In the aluminum housing it should avoid direct contact between the ground connection cable and the aluminum housing. It is therefore recommended to connect the safety ground cable, by placing it between the washer and the metal bracket on the related terminal or use an eyelet terminal crimped on the ground protection cable.
- ❑ The sensor, hardwired inputs and outputs are connected to the converter through terminal blocks located inside the converter.
- ❑ To locate the terminal block loosen the 4 screws on the terminal block cover. When the front cover is lifted, the terminal block is visible. The terminal block is the hardwire connection of the converter to external equipment, including the sensor.

The following pages give informations on the terminal block numbering, and the respective connecting of the sensor cables, and inputs/outputs.

INTERNAL LITHIUM BATTERY



- ❑ The internal lithium battery is rechargeable and should never be disconnected from the converter card when it is powered by the mains voltage. If this operation should be carried out it could irreparably damage the converter board.
- ❑ The internal lithium battery is recharged only when the converter is connected to the mains supply (LV, LLV, HV), battery charging condition, and not with the USB connection.
- ❑ During charging, the battery symbol appears on the MCP display and flashes blue; see "MEANING OF FLAGS" page 21 (the colors of the symbols can only be viewed in the virtual display of the MCP interface)
- ❑ When the battery charge falls below the minimum potential, battery low, the fixed red battery symbol appears; see "MEANING OF FLAGS" page 21 (the symbol colors can only be displayed in the virtual display of the MCP interface). Furthermore, in this charging condition below the minimum potential, the measurement does not start when the USB cable is connected.
- ❑ The thresholds that identify the condition of Battery low and battery charging are established by the system according to the use and settings assigned and therefore there is no fixed value..
- ❑ The two Battery low and battery charging icons can both be present as each one indicates a different condition.

GENERAL OPERATING NOTES

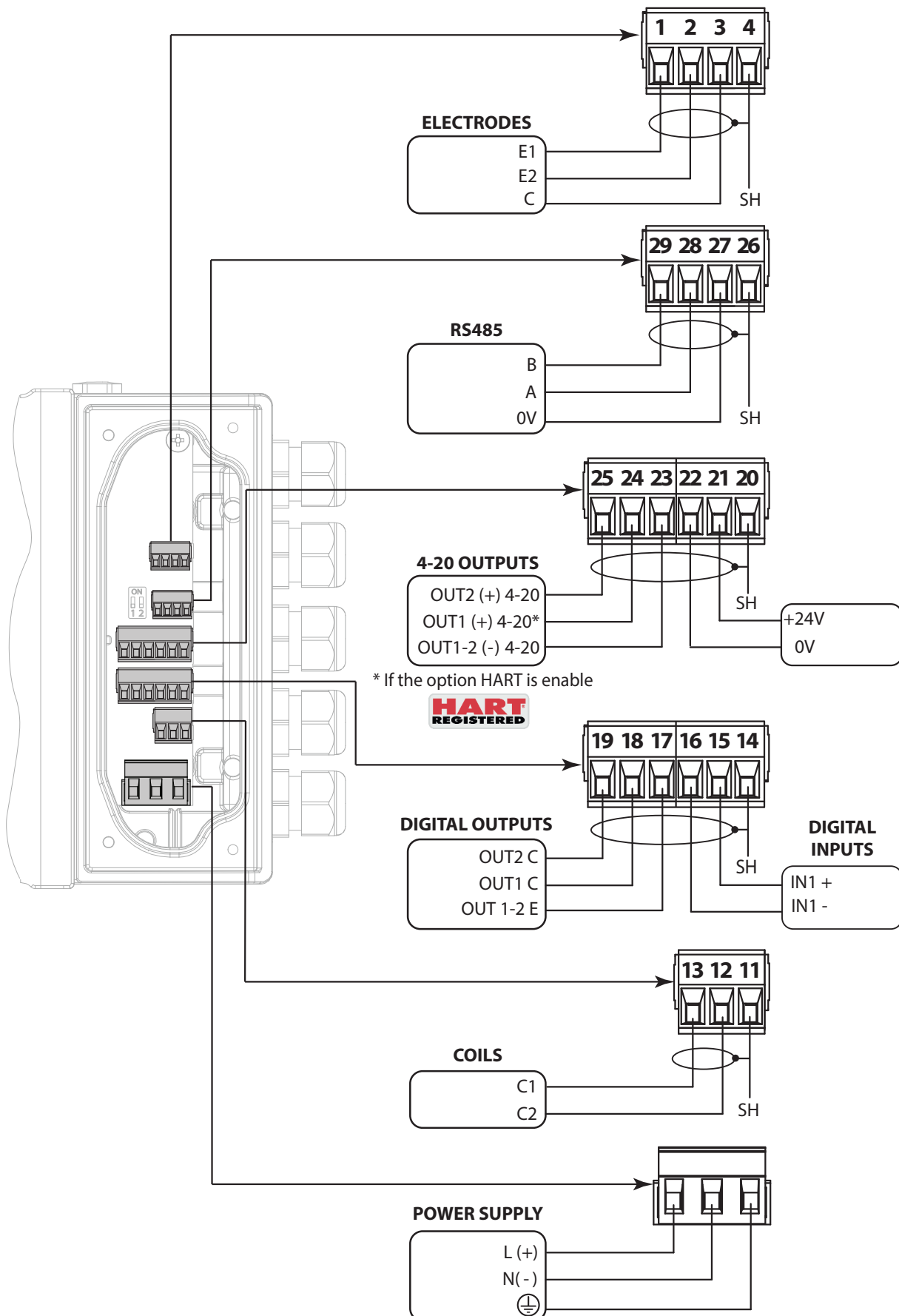
- ❑ If the mains voltage is disconnected from the converter with HV power supply, the flashing blue icon indicating the charging status can remain active for tens of seconds. This is due to the energy stored in the HV power supply and in the capacities of the circuit that discharge slowly.
- ❑ The battery voltage during charging does not rise immediately but gradually. This is related to the parameters set in the converter and detected by the controller chip.
- ❑ When the battery is physically disconnected the potential measured by the system is not true because the charging circuit checks the battery status by emitting impulses. These load the capacities in the circuit and the average value detected is not to be considered a reliable real value.
- ❑ The MCP command [SBCHS] indicates the percentage of charge and is not linear with respect to the battery voltage, however approximately it reads 0% with a voltage equal to about 3.2 V, while 100% occurs with about 4.1 V. These values are influenced by the temperature and the total operating time that worked the battery.

ELECTRICAL CONNECTION CONVERTER- SENSOR



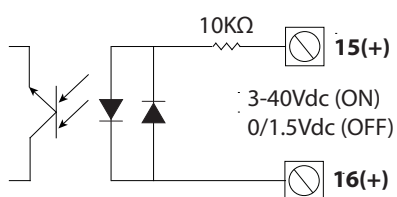
Sudden movements of the electrodes cable could introduce noise.

SH = SHIELD OF CABLE internally connected to ground.

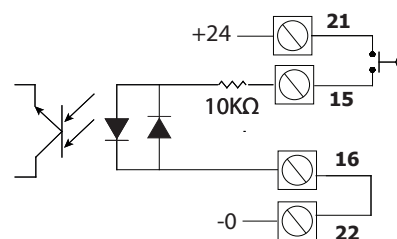


DIGITAL INPUT ON/OFF OPERATION

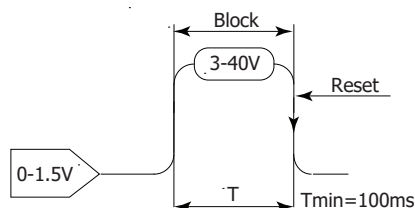
EXTERNAL POWER SUPPLY



INTERNAL POWER SUPPLY



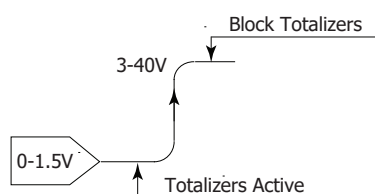
RESET TOTALIZERS



Necessary conditions for enabling the function

- ☐ POS. 6.1 see page 28 (T+; total direct positive set on)
- ☐ POS. 6.2 see page 28 (P+; partial direct positive set on)
- ☐ POS. 6.3 see page 28 (T-; total direct negative set on)
- ☐ POS. 6.4 see page 28 (P-; total direct negative set on)

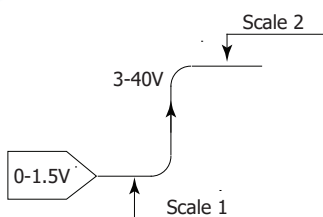
BLOCK TOTALIZERS



Necessary conditions for enabling the function

- ☐ POS. 6.5 see page 28 (Totalizer counting lock command set on)

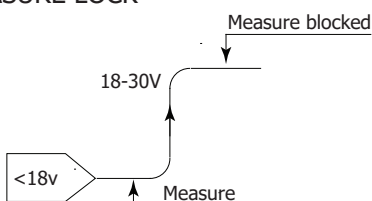
RANGE CHANGE



Necessary conditions for enabling the function

- ☐ POS. 6.8 see page 28 (Range change set on)

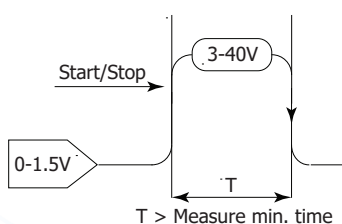
MEASURE LOCK



Necessary conditions for enabling the function

- ☐ POS. 6.6 see page 28 (Totalizer counting lock command set on)

BATCHING



Necessary conditions for enabling the function

- ☐ 7.1 Output 1 functions, Set to BATCHING

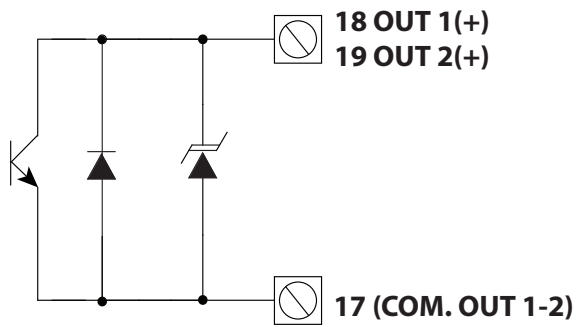


MUST BE
T > Tmin

MEASURE FREQ.	T MIN
50HZ	20ms
10HZ	100ms
5HZ	200ms
2HZ	500ms

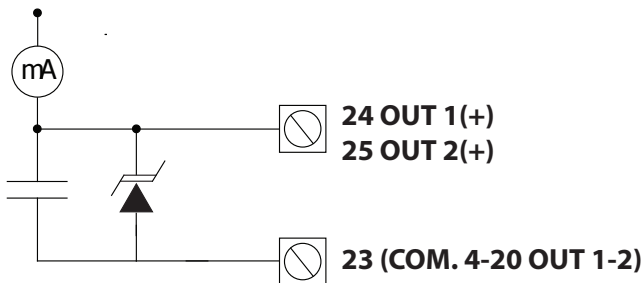
OUTPUTS WIRING

DIGITAL OUTPUTS



- ☐ Opto-insulated output with floating collector and emitter terminals freely connectable
- ☐ Maximum switching voltage: 40V ---
- ☐ Maximum switching current: 150mA
- ☐ Maximum saturation voltage between collector and emitter @150mA: 1.2V
- ☐ Maximum switching frequency (load on the collector or emitter, $R_L=470\Omega$, $V_{OUT}=24V$ ---): 1250Hz
- ☐ Maximum reverse current bearable on the input during and accidental polarity reversion (VEC): 100mA
- ☐ Insulation from other secondary circuits: 500V ---

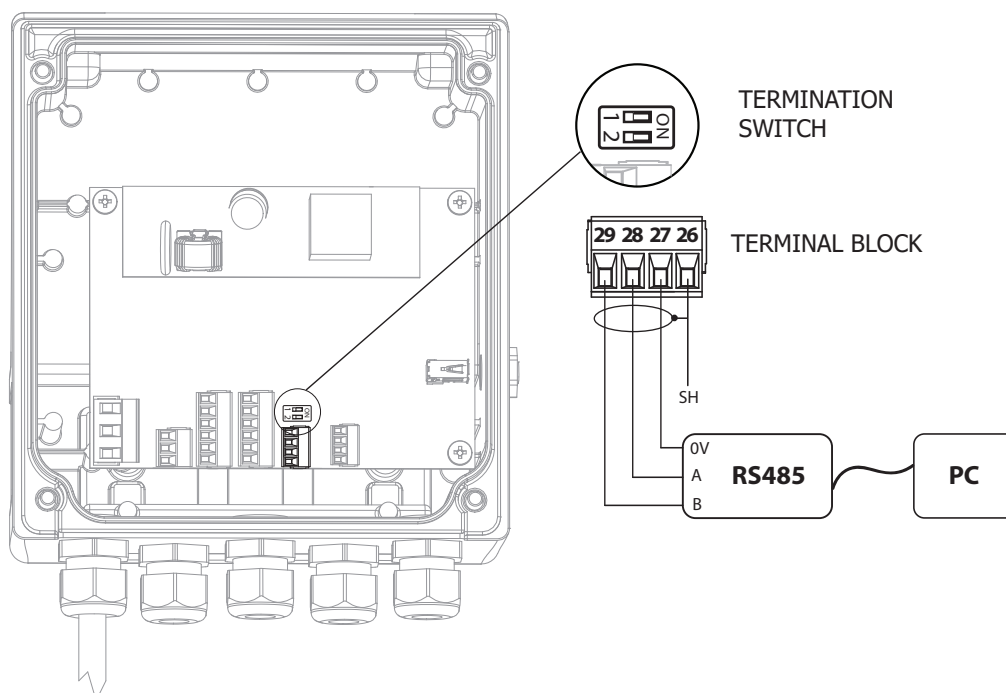
ANALOG OUTPUTS



- ☐ Opto-insulated output
- ☐ Maximum load: 1000Ω
- ☐ Maximum voltage without load: 27V
- ☐ Refresh frequency is the same of the sample frequency of the connected sensor
- ☐ Protected against persistent over voltages to maximum 30V

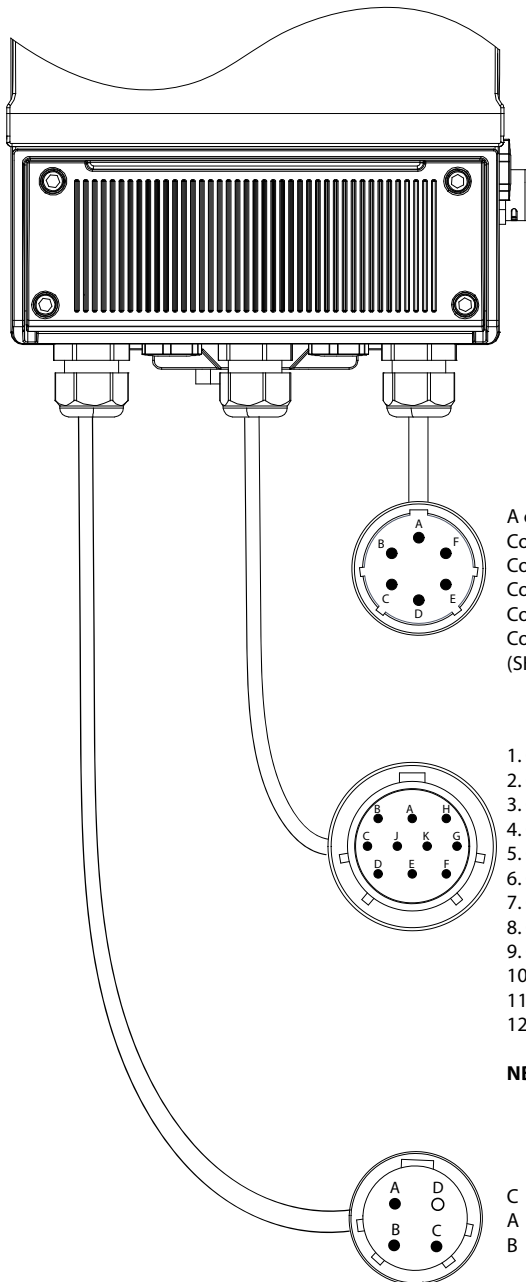
RS485 MODBUS MODULE (OPTIONAL)

Positioning to 'ON' the termination switches 1 and 2, a 120Ω resistance is activated in the RS485 circuit (see terminal block).



CONNECTORS MIL

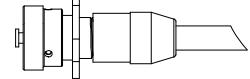
THE following are the links of the MIL connectors IP68



SENSOR SIGNALS

- A contact from terminal 1 of the converter (electrode 1)
- Contact F from terminal 2 of the converter (electrode 2)
- Contact And from terminal 3 of the converter (COM. Elec.)
- Contact B from terminal 13 of the converter (COIL 1)
- Contact C from terminal 12 of the converter (COIL 2)
- Contact D from terminal 4 and 11 of converter (SHIELD electrodes - COILS)

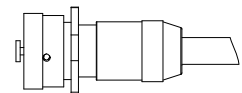
Connector
6 poles



INPUT/OUTPUT

- 1. Contact B from terminal 24 of the converter (Out1 4-20 +)
- 2. Contact A from terminal 25 of the converter (Out2 4-20 +)
- 3. Contact H from terminal 23 of the converter (Out1-2 4-20 -)
- 4. Contact C from terminal 21 of the converter (+ 24V)
- 5. Contact J from terminal 22 of the converter (0V)
- 6. Contact K from terminal 15 of the inverter (IN1 +)
- 7. Contact G from terminal 16 of the converter (IN1-)
- 8. Contact D from terminal 19 of the converter (C Out2)
- 9. Contact E from terminal 18 of the converter (Out1 C)
- 10. Contact F from terminal 17 of the converter (Out1-2 E)
- 11. Contact C from terminal 28 of the converter (RS485 A)
- 12. Contact J from terminal 29 of the converter (RS485 B)

Connector
10 poles

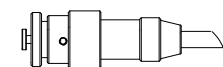


NB: the connections 4 and 5 exclude the use connections 11 and 12.

POWER SUPPLY

- C from terminal L power converter (+ dc)
- A from terminal N power converter (-dc)
- B from terminal GROUND Power Converter

Connector
4 poles



NOTE: Military Connector 6 poles for sensor converter only provided in the separate version of the converter.

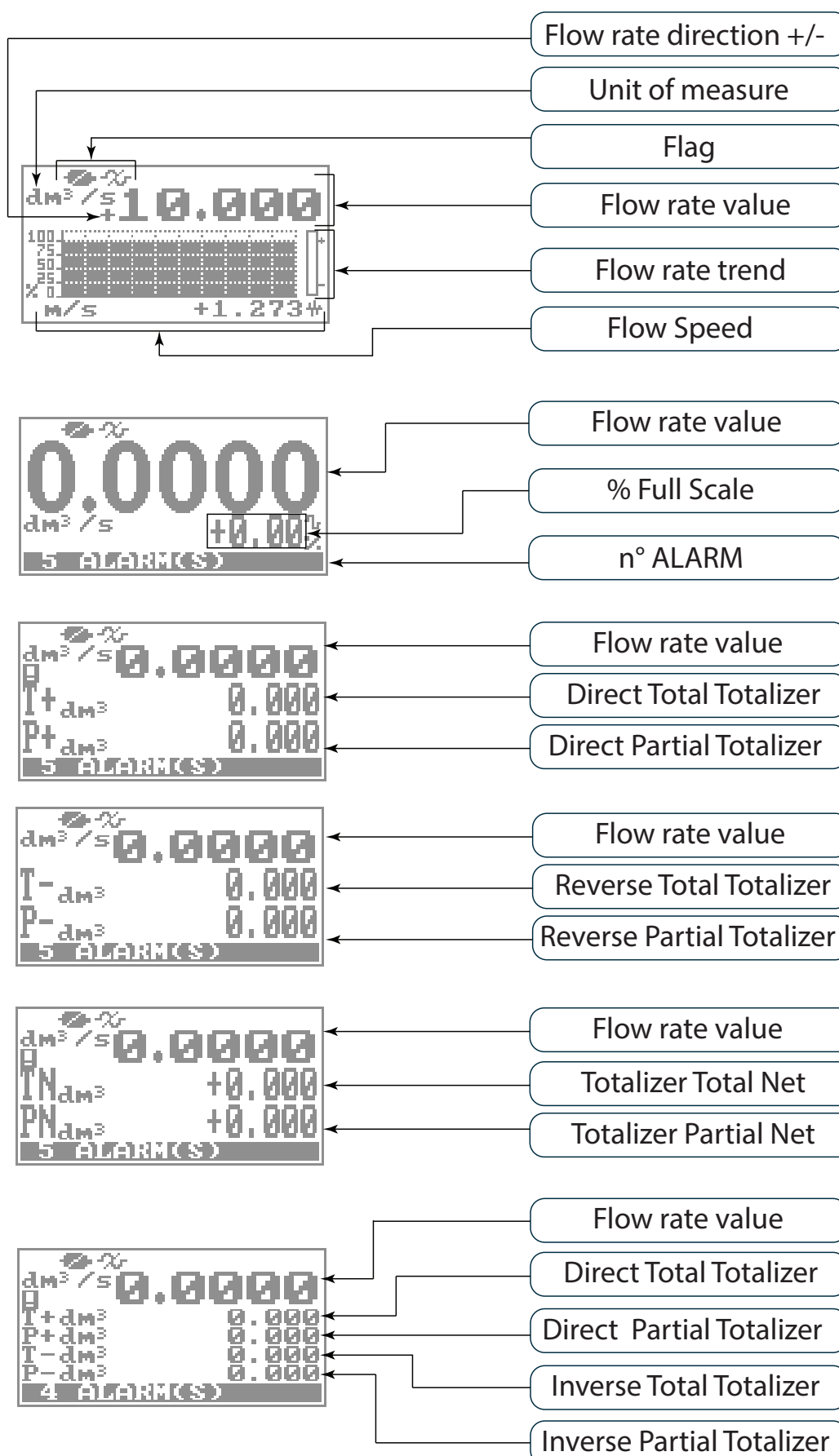
START VISUALIZATION PAGES



The direct exposure of the converter to the solar rays, could damage the liquid crystal display. The visualization pages can be change according to instrument's setup.

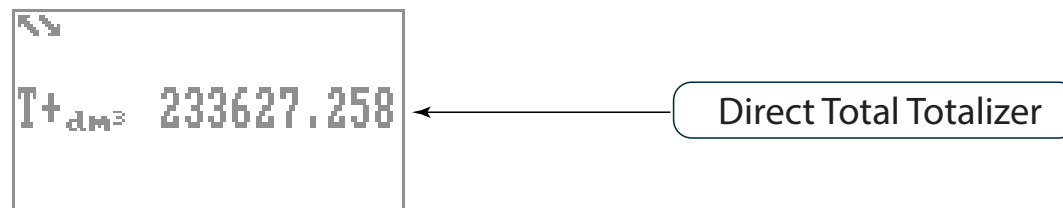
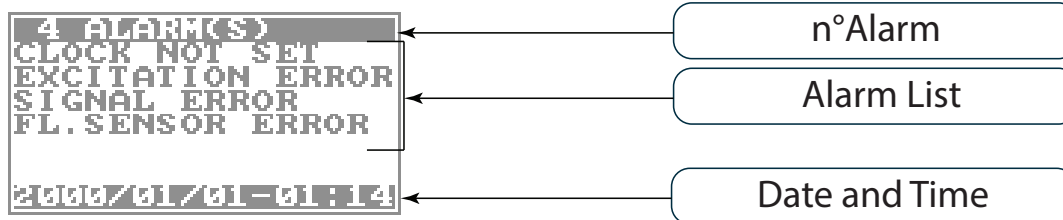
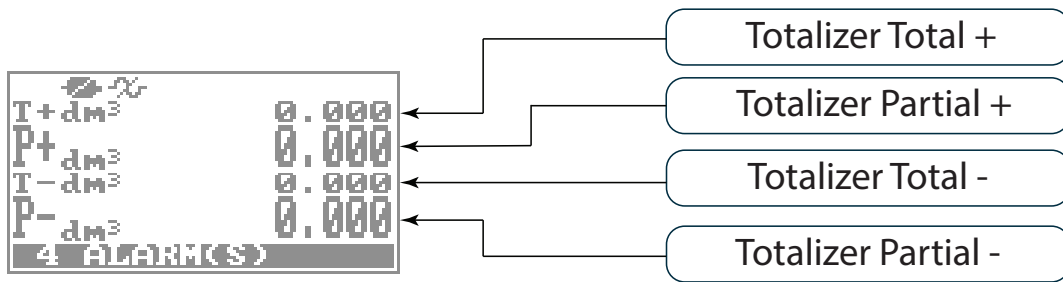


Push to change visualization





Push to change visualization

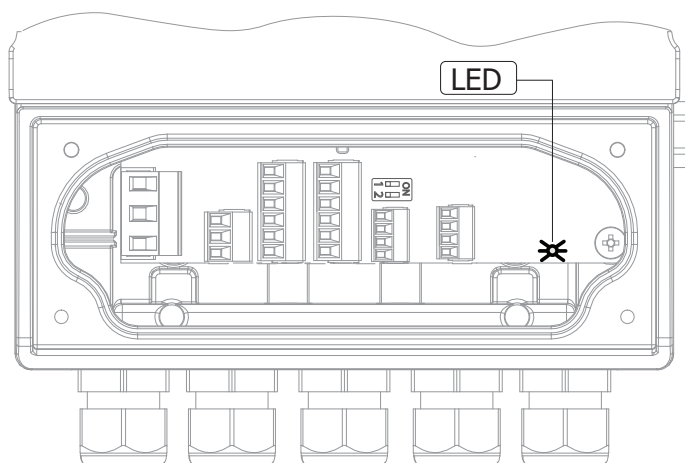


[The visualisation page about batching is described in the dedicated section "BATCHING" pag45](#)

MEANING OF FLAGS

FLAG	DESCRIPTION	FLAG	DESCRIPTION
	EMPTY PIPE		MIN FLOW ALARM
	FILE UPLOAD		MAX FLOW ALARM
	FILE DOWNLOAD		VIDEO TERMINAL CONNECTED
	BATTERY RECHARGE (FLASHING) LOW BATTERY (FIXED)		FLOW RATE OVERFLOW
	FLOW RATE SIMULATION (FLASHING)		PULSE 1 OVERFLOW
	CALIBRATION (FLASHING)		PULSE 2 OVERFLOW
	GENERIC ALARM (FLASHING)		POWERED DEVICE WITH ONE CHARGERS BATTERY (MID-DIRECTIVE) oppure BATCHING IN PROGRESS
	GENERAL ALARM ONLY ON PHYSICAL DISPLAY (FLASHING)		
	SIGNAL ERROR		
	EXCITATION ERROR		

MEANING OF LED COLORS



LED Red: Alarm signal

LED Blue: Usb communication enable

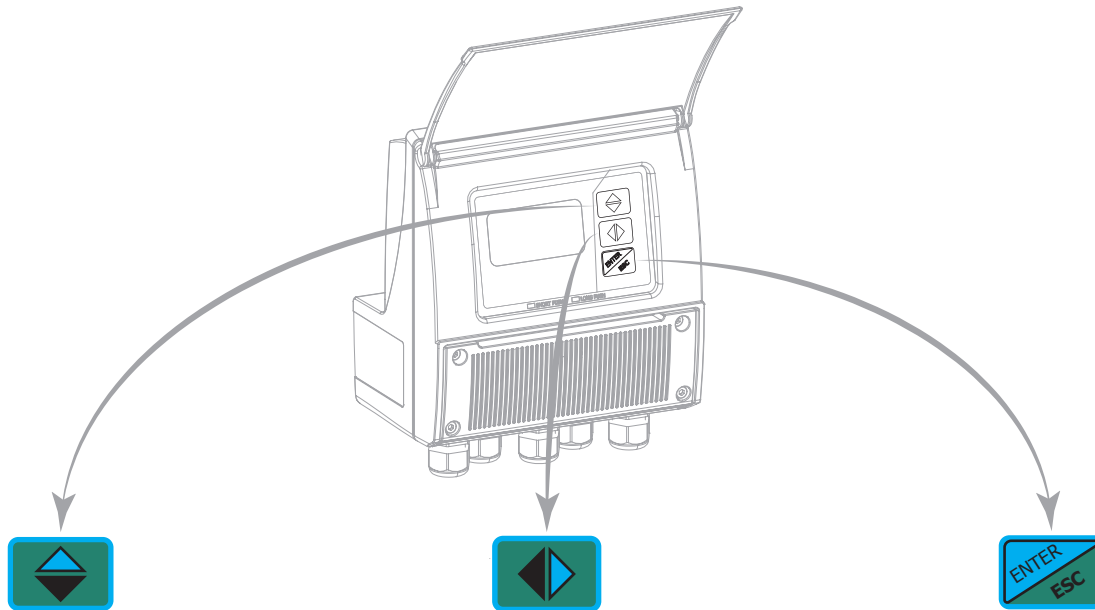
LED Green: Functioning system correctly

ACCESS TO THE CONFIGURATION MENU

The configuration can be done in two different ways:

- ☐ By keypad of converter
- ☐ By MCP interface (Virtual display of instrument)

ACCESS VIA KEYPAD



SHORT PRESSING (< 1 SECOND):

Increases the numeric figure or the parameter selected by the cursor
Returns to the previous subject on the menu.

LONG PRESSING (> 1 SECOND):

Decreases the numeric figure or the parameter selected by the cursor.
Proceeds to the next subject on the menu.

SHORT PRESSING (< 1 SECOND):

Moves/positions the cursor rightward on the input field. Proceeds to the following subject of the menu. Change the display of the process data

LONG PRESSING (> 1 SECOND):

Moves/positions the cursor leftward on the input field. Returns to the previous subject on the menu

SHORT PRESSING (< 1 SECOND):

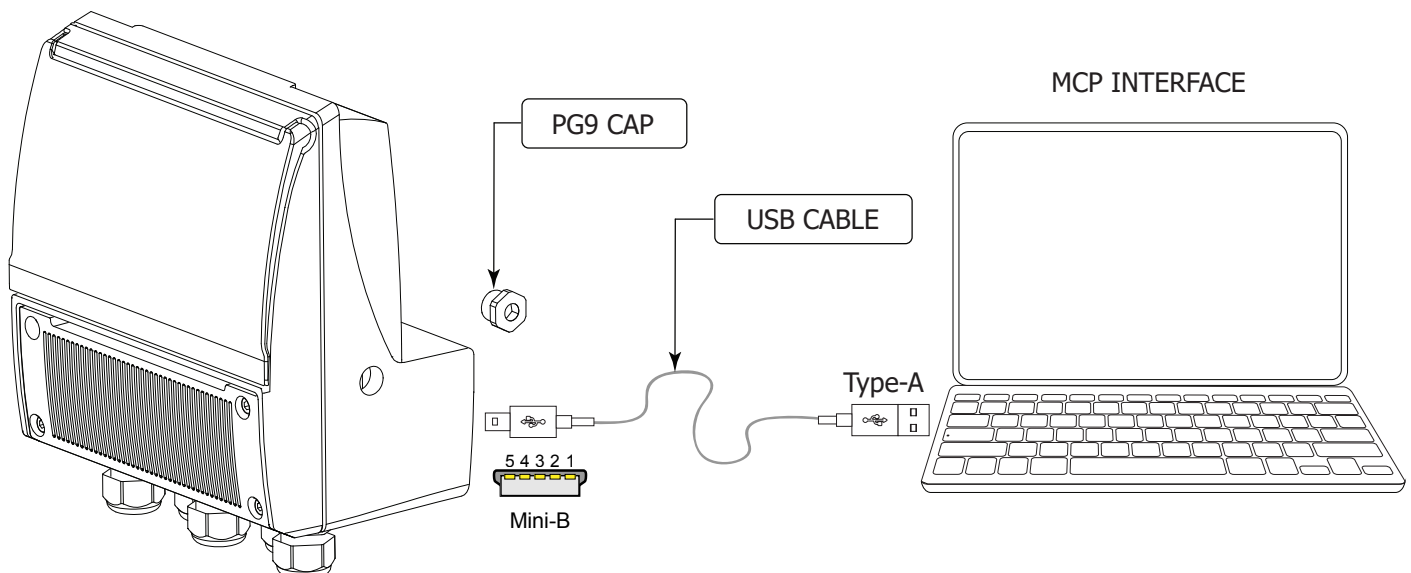
Enter /leave the selected function
Enables the main menu for the instrument configuration Cancels the selected function under progress

LONG PRESSING (> 1 SECOND):

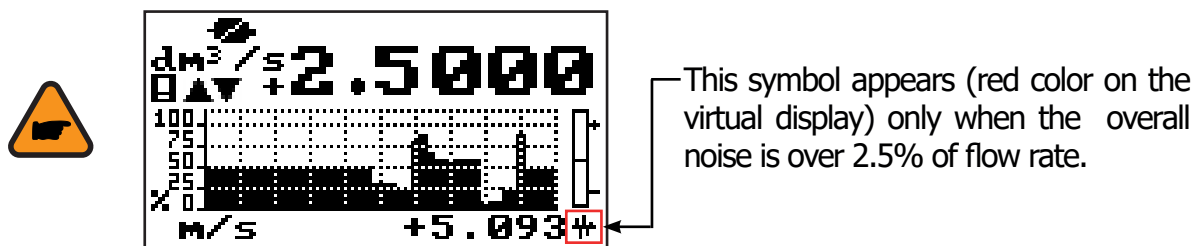
Leaves the current menu
Enables the totalizer reset request (when enabled) Confirms the selected function.

ACCESS VIA MCP INTERFACE (VIRTUAL DISPLAY)

MCP is a Windows® software that allows to set all the converter functions and personalize the menu. The MCP program is required for the blind version of the converter. To use MCP interface consult the relevant user manual.



FLOW RATE VISUALIZATION



The MV 210 can show a 5 digits display for flow rate units; this mean the maximum flow rate value that can be represented on the display is 99999 (no matter the positioning of the decimal point). The minimum is 0.0025. The representable measure unit depends on sensor flow rate and diameter; the permitted units are those, that permits the instrument full scale value not exceeding 99999.

Example for DN 300, Full scale value: 3m/s:

❑ **PERMITTED measure unit** (example): l/s (216.00); m³/h (777.60); m³/s (0.2160)

❑ **NOT PERMITTED measure unit** (example): l/h (777600)

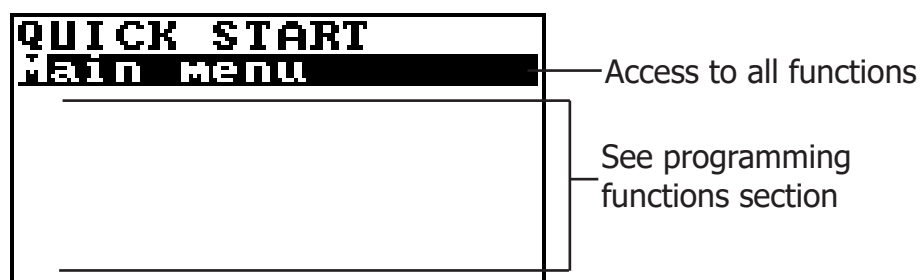
FLOW RATE ALERT



This FLAG becomes active when there is a flow variation (flow rate not stable).

QUICK START MENU

The QUICK START MENU allows to user immediate access to some of the most commonly used functions; through MCP software it possible customize this menu to make it suitable for the specific application.



The user has immediate access to the Quick Start menu when the converter is powered up by pressing the Enter key. If access to the quick start menu does not occur, then it could be disabled using the function "9.11" page 29 .

CONVERTER ACCESS CODE

The access for programming the instrument is regulated by six access levels logically grouped. Every level is protected by a different code.

❑ Access Level 1-2-3-4 Freely programmable by user

ACCESS CODE SET : MENU 13 SYSTEM

```

SYSTEM
Default.saving= ON
Time zone=h+01.00
2016/04/04-16:07
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
Restr.access= ON
010.011.012.013
010.011.012.014
255.255.255.000
KI= 0.96469
KS= 1.00000
KR= 1.00000
DAC1 4mA= 02460
DAC1 20mA= 11050
DAC2 4mA= 02460
DAC2 20mA= 11050
Stand-by
FW update
13-System
  
```

```

SYSTEM
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
0499999999
  
```

The CODE is Settable by keyboard or MCP interface.
Depending on the level of access different display functions will be visible. (See section "FUNCTIONS DESCRIPTION" page 32)
These access levels interact with the "Restricted access"

RESTRICTED ACCESS SET : MENU 13 SYSTEM

```

SYSTEM
L1 code=*****
L2 code=*****
L3 code=*****
L4 code=*****
L5 code=*****
L6 code=*****
Restr.access=OFF
  
```

Settable Values

ON

OFF

Restrict = ON: Access permitted only to functions provided for a specific level;
Example: If the operator has a code of access level 3, after having set it, he can change only the functions with level 3 access.

Restict = OFF: It enables to change functions for the selected level and ALL the functions with lower access level.

Example: If the operator has the code of level 3, after having set it, he can change all the functions at level 3 and those at lower levels.

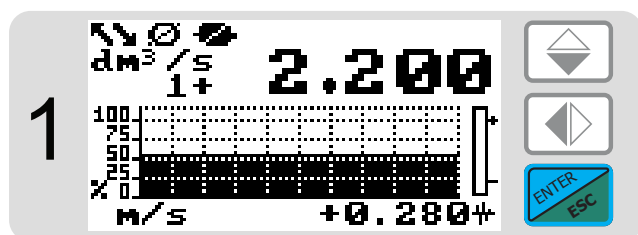
* WARNING: take careful note of the customized code, since there is no way for the user to retrieve or reset it if lost.

Factory preset access codes:

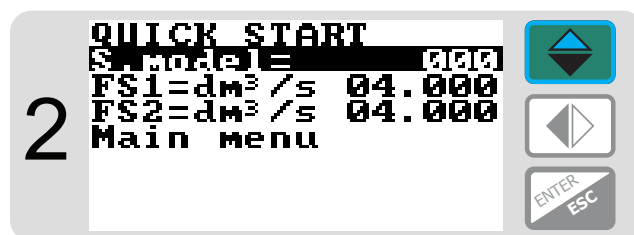
- ❑ L1: 10000000
- ❑ L2: 20000000
- ❑ L3: 30000000
- ❑ L4: 40000000

The following example shows how to change the Full scale by Quick Start menu; the second illustrates how to change the function by the Main menu.

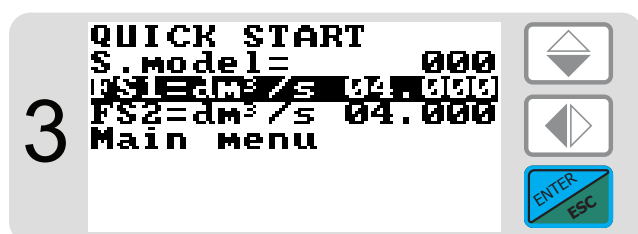
EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Quick start menu"



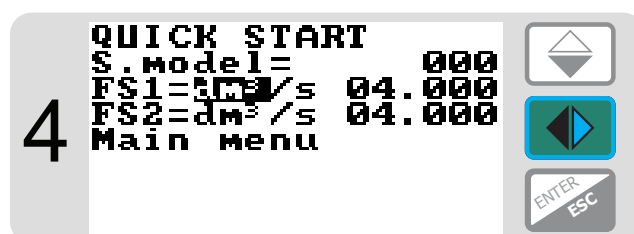
Press the ENTER button to access the Quick Start menu



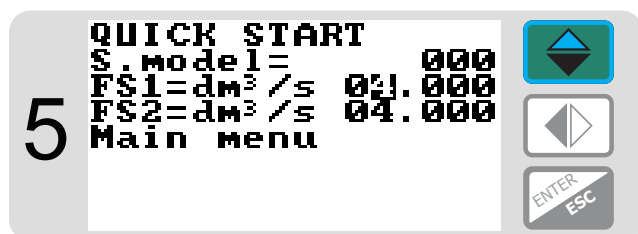
Select this function in the list to be edited



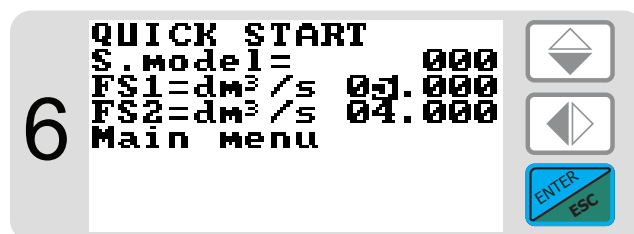
Press the ENTER button to select the function.



Select the value to be changed



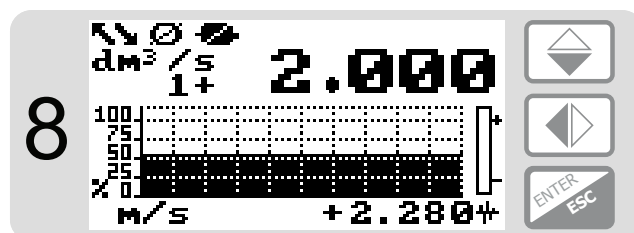
Change the value



Confirm the new value

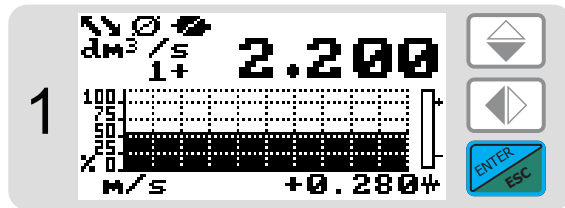


Long Push

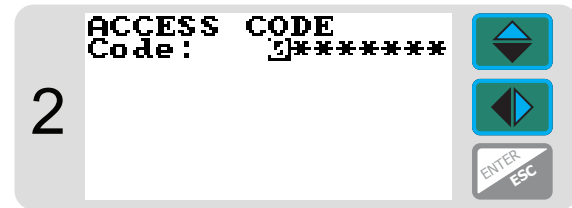


Main Page

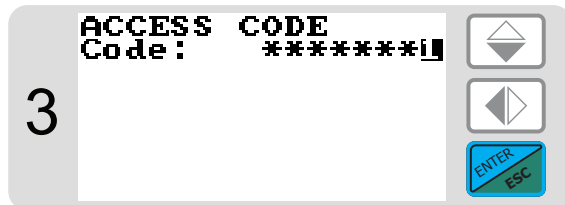
EXAMPLE: modifying the full scale value from 4dm³/s to 5dm³/s, from the "Main Menu" (quick start menu enabled)



Press the ENTER button to access the Quick Start menu



Press arrow keys to select the cell in which to insert the number of the access code.



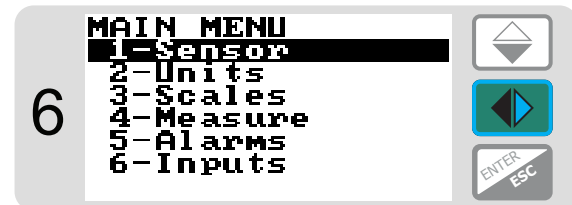
Press ENTER button to confirm value.



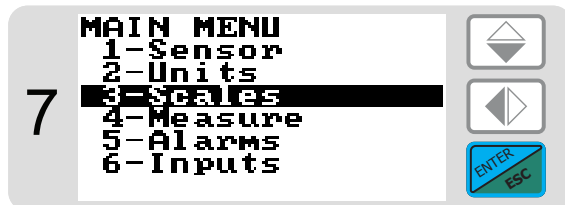
Select "Main Menu"



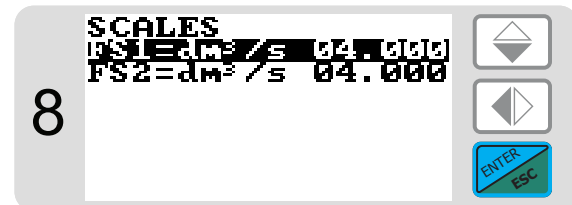
Press the ENTER button to access the Main Menu



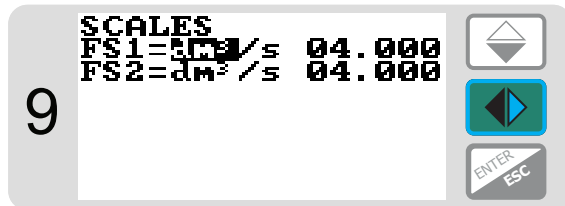
Select function



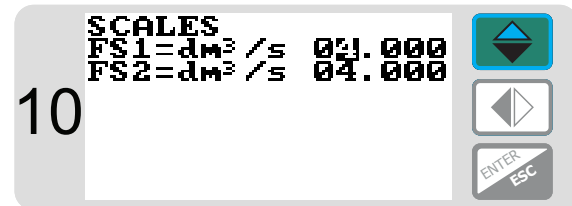
Press the ENTER button to access the "Scale Menu"



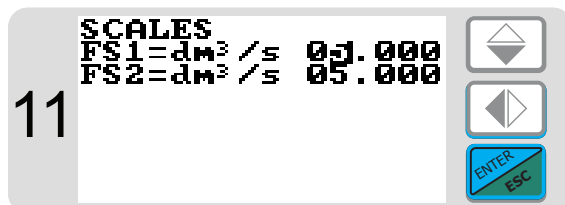
Press the ENTER button to access the "Fs1"



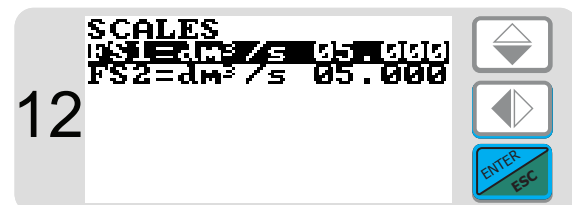
Select the value to be changed



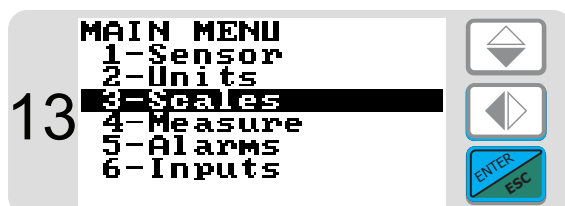
Change the value



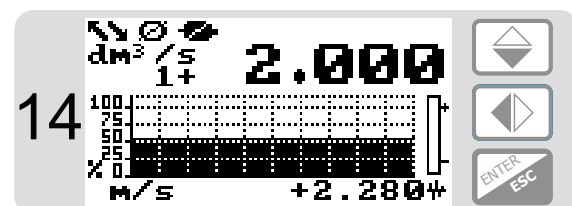
Press the ENTER button Confirm the new value



Press Esc



Press Esc



Main page

FUNCTIONS MENU

The main menu is selected from the Quick start menu by pressing the  key and entering the access code; enter the access code if required. Note: Some functions are displayed only with other functions active, or with optional modules.

SENSOR

MAIN MENU			
1	Sensor		
2	Sensor		
3	Sensor		
4	Sensor		
5	Sensor		
6	Sensor		
7	Sensor		
8	Sensor		
9	Sensor		
10	Sensor		
11	Sensor		
12	Sensor		
13	Sensor		
	Sensor		
	S. model =	0	1.1 Sensors model: Enter the first two characters of the serial number of the sensor
	Lining =	UNSPEC.	1.2 Flow sensor lining material type
	S. type =	FULL BORE	1.3 Type of sensor: fullbore or insertion
	u.type =	METRIC	1.4 Type of measure units for sensor parameter: metric or imperial
	Diam	mm 00025.0	1.5 Insert ND of sensor (0-2500)
	HA =	+00.9637	1.6 Calibration data of sensor visualized on sensor's label
	HA- =	-04.4904	1.7 Calibration data of sensor for negative flow
	HZ =	+0000000	1.8 Sensor coefficient KZ (zero point)
	HD =	+0000000	1.9 Sensor coefficient KD
	Ins. position =	0	1.10 Insertion position
	HP DinaWmic =	OFF	1.11 KP dynamic, coefficient for insertion
	Ri =	01.8727	1.12 Sensor coefficient Ki
	Kp =	01.0000	1.13 Sensor coefficient Kp
	KC =	1.0000	1.14 Sensor coefficient KC
	C.curr =	025.0	1.15 CWV volume total, decimal point position
	C.Reg.PB =	004	1.16 Current regulator proportional band
	C.Reg.DR =	008	1.17 Current regulator derivation constant
	S. Freq. = Hz	50	1.18 Measure sampling frequency
	Preamplif.	OFF	1.19 Enables the preamplifier
	E.P Detect =	ON	1.20 Enables the empty pipe detection feature
	R max = kohm	0500	1.21 Empty pipe detection threshold
	El. Cleaning =	OFF	1.22 Electrode cleaning
	S. cable =	m 000	1.23 Sensor connecting cable length
	S. err. delay =	010	1.24 Signal error delay (n. sample)
	Sens. verify =	OFF	1.25 Automatic sensor verify enable
	Zero point cal.		1.26 Pipe hydraulic zero calibration
	KL	00.0000000	1.27 Linearization coefficient

UNITS

MAIN MENU			
1	Sensor		
2	Units		
3	Scales		
4	Units		
5	Units		
6	Units		
7	Units		
8	Units		
9	Units		
10	Units		
11	Units		
12	Units		
13	Units		
	Units		
	Diam.	mm	2.1 Nominal diameter measure unit
	S.cable	m	2.2 Cable length on separate version
	FR.unit	METRIC	2.3 Flowrate type measure unit: metric or not metric
	PI1 unit	METRIC	2.4 Pulse 1 type measure unit: metric or not metric
	PI2 unit	METRIC	2.5 Pulse 2 type measure unit: metric or not metric
	T+ unit	METRIC	2.6 Total direct totalizer measure unit type: metric or not metric
	T+ unit	g	2.7 Total direct totalizer measure unit
	T+ D.P	3	2.8 Total direct totalizer decimal point position
	P+ unit	METRIC	2.9 Partial direct totalizer measure unit type: metric or not metric
	P+ unit	g	2.10 Partial direct totalizer measure unit
	P+ D.P	3	2.11 Partial direct totalizer decimal point position
	T- unit	METRIC	2.12 Total reverse totalizer measure unit type: metric or not metric
	T- unit	g	2.13 Total reverse totalizer measure unit
	T- D.P	3	2.14 Total reverse totalizer decimal point position
	P- unit	METRIC	2.15 Partial reverse totalizer measure unit type: metric or not metric
	P- unit	g	2.16 Partial reverse totalizer measure unit
	P- D.P	3	2.17 Partial reverse totalizer decimal point position
	Temp.unit	°C	2.18 Temperature measure unit
	Mass units	ON	2.19 Enable/disable the selection of mass units on full scale set
	Sg=kg/dm3	1.0000	2.20 Specific gravity coefficient

SCALES

MAIN MENU		
1	Sensor	
2	Units	
3	Scales	
4	Measure	
5	Alarms	
6	Inputs	
7	Outputs	
8	Settings	
9	Help	
10	Exit	
11	Back	
12	Forward	
13	Cancel	

SCALES		
1	FS1 g/s	4908.7
2	B.qty dm3	1.00000
3	FS2 g/s	4908.7
4	Pls1=g	1000.00
5	Tpls1=ms	0050.0
6	Pls2=g	1000.00
7	Tpls2=ms	0050.0
8	Frq1=Hz	1000.00
9	Frq2=Hz	1000.00

3.1	Full scale flow rate 1
3.2	Batching quantity
3.3	Full scale flow rate 2
3.4	Pulse value on channel 1
3.5	Duration of the pulse generated on channel 1
3.6	Pulse value on channel 2
3.7	Duration of the pulse generated on channel 2
3.8	Full scale frequency for channel 1 (0.1Hz-1000.0Hz)
3.9	Full scale frequency for channel 2 (0.1Hz-1000.0Hz)

MEASURES

MAIN MENU		
1	Sensor	
2	Units	
3	Scales	
4	Measure	
5	Alarms	
6	Inputs	
7	Outputs	
8	Settings	
9	Help	
10	Exit	
11	Back	
12	Forward	
13	Cancel	

MEASURES		
1	Damping	SMART
2	Cut-off= %	00.1
3	DT Min.	ON
4	T1HC enable	ON
5	T1HC	OFF

4.1	Measure filter
4.2	Low flow zero threshold: 0-25% of full scale value
4.3	Automatic calibration verify
4.4	Automatic change of measurement range
4.5	High immunity inputs

ALARMS

MAIN MENU		
1	Sensor	
2	Units	
3	Scales	
4	Measure	
5	Alarms	
6	Inputs	
7	Outputs	
8	Settings	
9	Help	
10	Exit	
11	Back	
12	Forward	
13	Cancel	

ALARMS		
1	Max+ = dm3/s	OFF
2	Max- = dm3/s	OFF
3	Min+ = dm3/s	OFF
4	Min- = dm3/s	OFF
5	Hysteresis=%	03
6	mA v.alarm=%	000
7	Hz v.alarm=%	000

5.1	Maximum value alarm set for direct flow rate
5.2	Maximum value alarm set for reverse flow rate
5.3	Minimum value alarm set for direct flow rate
5.4	Minimum value alarm set for reverse flow rate
5.5	Hysteresis threshold set for the minimum and maximum flow rate alarms
5.6	Current output value in case of failure
5.7	Frequency output value in case of alarms

INPUTS

MAIN MENU		
1	Sensor	
2	Units	
3	Scales	
4	Measure	
5	Alarms	
6	Inputs	
7	Outputs	
8	Settings	
9	Help	
10	Exit	
11	Back	
12	Forward	
13	Cancel	

INPUTS		
1	T+ reset	OFF
2	P+ reset	OFF
3	T- reset	OFF
4	P- reset	OFF
5	Count lock	OFF
6	Meas.lock	OFF
7	Calibration	OFF
8	Range change	OFF

6.1	Total direct (positive) flow totalizer reset enable
6.2	Partial direct (positive) flow totalizer reset enable
6.3	Total reverse (negative) flow totalizer reset enable
6.4	Partial reverse (negative) flow totalizer reset enable
6.5	Totalizer counting lock command
6.6	Measure zero lock command
6.7	Calibration external command
6.8	Range change external command

OUTPUTS

MAIN MENU	
1-Sensor	
2-Units	
3-Scales	
4-Measure	
5-Alarms	
6-Inputs	
7-Outputs	
8-Communication	
9-Display	
10-Data logger	
11-Functions	
12-Settings	
13-Exit	
OUTPUTS	
10- Out1	PULSES+
11- Out2	PULSES-
12- Out mA1	4_22 +/-
13- Out mA2	4_22 +/-
A1S	4.9087
A2S	4.9087

7.1	Output 1 functions
7.2	Output 2 functions
7.3	Choice of the function and the range of current output n.1
7.4	Choice of the function and the range of current output n.2
7.5	Full Scale value for analog out1
7.6	Full Scale value for analog out2

COMMUN.

MAIN MENU	
1-Sensor	
2-Units	
3-Scales	
4-Measure	
5-Alarms	
6-Inputs	
7-Outputs	
8-Communication	
9-Display	
10-Data logger	
11-Functions	
12-Settings	
13-Exit	
COMMUNICATION	
10- HART pr.	05
11- HART O. C.	ON
12- Dev. Addr	001
13- Speed=bps	9600
Parity=	NO
Delay=ms	00
C. timeout	2

8.1	HART packet byte preambles
8.2	HART bus output control
8.3	Device communication address number
8.4	MODBUS link speed
8.5	MODBUS link parity
8.6	MODBUS reply delay
8.7	Maximum delay between chars (frames) MODBUS



DISPLAY

MAIN MENU	
1-Sensor	
2-Units	
3-Scales	
4-Measure	
5-Alarms	
6-Inputs	
7-Outputs	
8-Communication	
9-Display	
10-Data logger	
11-Functions	
12-Settings	
13-Exit	
DISPLAY	
Language	EN
Contrast	5
Disp.time=s	020
D.rate=Hz	5
Disp. Fn.	1
Disp.lock	ON
Part. Tot	ON
Neg. Tot.	ON
Net tot.	ON
Disp.date	ON
Quick start	ON

9.1	Choice of the language
9.2	Display contrast
9.3	Display/keyboard inactivity time
9.4	Display updating frequency: 1-2-5-10 Hz
9.5	Display function number
9.6	Display function selection lock
9.7	Partial totalizer enable
9.8	Negative totalizer enable
9.9	Net totalizer enable
9.10	Time and date display enable
9.11	Quick start menu visualization

DATA LOGGER

DATA LOGGER		
D.logger en.	ON	10.1 Data logger enabling
Meas. units	ON	10.2 Measure units recording enable
Field separat.	;	10.3 Field separator character
Decimal separat.	.	10.4 Decimal separator character
Interv.	01:01:00	10.5 Sampling interval
Log T+	ON	10.6 Enable logging of total direct totalizer
Log P+	ON	10.7 Enable logging of partial direct totalizer
Log T-	ON	10.8 Enable logging of total reverse totalizer
Log P-	ON	10.9 Enable logging of partial reverse totalizer
Log TN	ON	10.10 Enable logging of total net totalizer
Log PN	ON	10.11 Enable logging of partial net totalizer
Log Q (UM)	ON	10.12 Enable logging of flow rate in measure unit
Log Q (%)	ON	10.13 Enable logging of flow rate in percentage
Log AL.EV	ON	10.14 Enable logging of alarm events
Log STR	ON	10.15 Enable logging of sensor test results
Log BTS	ON	10.16 Enable logging of board temperature
Log IBV	ON	10.17 Enable logging of internal board voltage
Log EDC	ON	10.18 Enable logging of electrodes DC voltage
Log EAC	ON	10.19 Enable logging of electrodes AC voltage
Log EIZ	ON	10.20 Enable logging of electrodes impedance
Log SCV	ON	10.21 Enable logging of sensor coils value
4-Outputs		
8-Communication		
9-Display		
10-Data logger		
11-Functions		
12-Diagnostic		
13-System		

FUNCTIONS

FUNCTIONS		
T+ reset		11.1 Execute immediate reset of total direct totalizer
P+ reset		11.2 Execute immediate reset of partial direct totalizer
T- reset		11.3 Execute immediate reset of total reverse totalizer
P- reset		11.4 Execute immediate reset of partial reverse totalizer
Load Sens. F. def		11.5 Load sensor factory default
Load Conv. F. def		11.6 Load converter factory default
Save Sens. F. def		11.7 Save sensor factory default values
Save Conv. F. def		11.8 Save converter factory default values
Calibration		11.9 Execute immediate internal circuit calibration
9-Display		
10-Data logger		
11-Functions		
12-Diagnostic		
13-System		

DIAGNOSTIC

DIAGNOSTIC		
Self test		12.1 Self test diagnostic function
Test display		12.2 Function tests physical display
Sens. verify		12.3 Sensor verify diagnostic function
Flow sim. =	ON	12.4 Flow rate simulation enabling
Display measures		12.5 Display internal measured value
Disp. Comm. Vars		12.6 Display comm. diagnostic values
Display graphs		12.7 Display measure as graphs
Gen. sens. set		12.8 Generic sensor parameters set
SD card info		12.9 Sd card status informations
Firmware info		12.10 Firmware version/revision
S/N=	999001	12.11 Board serial number
WT=	002:21:00 : 22	12.12 Total working time
4-Outputs		
9-Display		
10-Data logger		
11-Functions		
12-Diagnostic		
13-System		

SYSTEM

SYSTEM		
Dayl. Saving =	ON	13.1 Daylight saving time change
Time zone =	h+01.00	13.2 Localized time zone
	2016/04/04-16:07	13.3 System date and time
L1 code =	*****	13.4 Access level 1 code
L2 code =	*****	13.5 Access level 2 code
L3 code =	*****	13.6 Access level 3 code
L4 code =	*****	13.7 Access level 4 code
L5 code =	*****	13.8 Access level 5 code
L6 code =	*****	13.9 Access level 6 code
Restr. Access=	ON	13.10 Restricted access level
	010.011.012.013	13.11 Device IP network address
	010.011.012.014	13.12 Client IP network address
	255.255.255.000	13.13 Network mask
HT	0.96469	13.14 Calibration coefficient KT
HS	1.00000	13.15 Calibration coefficient KF
HR	1.00000	13.16 Calibration coefficient KR
DAC1	(°C)	13.17 DAC1 out 4mA calibration point
DAC1	(°C)	13.18 DAC1 out 20mA calibration point
DAC2	661	13.19 DAC2 out 4mA calibration point
DAC2	3327	13.20 DAC2 out 20mA calibration point
Stand-by	3453	13.21 Stand-by
FW update	14718	13.22 firmware update
9-Display		
10-Data logger		
11-Functions		
12-Diagnostic		
13-System		

FUNCTIONS DESCRIPTION



Here below the explanation on how the rows of menu are described.

Menu visualized on the converter (from 1 to 13)
MENU 1 - SENSOR

(POS. 1.1) <i>Sensor Model</i>	[S. model xxx]	AL4	[SMODL]
		Access level	MCP command
		Convert request1	
		Synthetic description of the function	

The following picture describes where to find the name of the MCP functions in MCP-software. More info see MCP manual.

The screenshot displays the McpProg software interface. On the left, a 'Function list' window shows a tree structure with 'Sensor' selected. The main window shows a large digital display with '0.00' and various units. Below the display, there are buttons for 'MCP' and 'OP1PV'. A callout box points to the 'MCP' button, indicating it is the 'Line editor for the insertion and execution of MCP commands.' Another callout box points to the 'OP1PV' button, indicating it is the 'MCP command'.

MENU 1 - SENSOR

(POS. 1.1) <i>Sensor model</i>	[S. model xxx]	AL4	[SMODL]
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Enter the first two characters of the serial number of the sensor as on the sensor label.

(POS. 1.2) <i>Lining Type</i>	[Lining= UNSPEC.]	AL4	[LIMAT]
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Flow sensor lining material type. (PFA; PU-TDI; ALON; PEEK; HR; PP; PA-11; PTFE-HT; PTFE)

(POS. 1.3) <i>Type of sensor</i>	[S. type= FULL BORE]	AL4	[STYPE]
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Select the sensor type of full-bore or insertion.

(POS. 1.4) <i>Unit type</i>	[U.type= METRIC]	AL4	[SUTYP]
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Select type of measure unit of sensor's parameter. Values metric or imperial (inch).

(POS. 1.5) <i>Diameter</i>	[Diam.= mm xxx]	AL4	[PDIMV]
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Select the nominal diameter of the sensor (0-2500). ND is written on the sensor label.

(POS. 1.6) <i>KA</i>	[KA= + xx.xxx]	AL4	[CFFKA]
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KA factor: calibration coefficient

(POS. 1.7) <i>KA -</i>	[KA= - xx.xxx]	AL4	[CFKAN]
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KA factor: calibration coefficient for negative flow. This function is showed only if at least 1 negative KL value is set.

(POS. 1.8) <i>KZ</i>	[KZ= +/- xxxxx]	AL4	[CFFKZ]
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Calibration Factor. KZ

(POS. 1.9) <i>KD</i>	[KD= +/- xxxxx]	AL4	[CFFKD]
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Calibration Dynamic Factor.

(POS. 1.10) <i>Insertion position</i>	[Ins.position= x]	AL4	[SIPOS]
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This function is active with POS.1.3 on "Insertion". See the insertion sensor manual for more details

(POS. 1.11) <i>KP dynamic</i>	[KP dynamic= ON/OFF]	AL4	[CFFKP]
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This function is active with POS.POS. 1.3 see page 27 set on insertion. See manual of insertion sensor for more details

(POS. 1.12) <i>Ki</i>	[Ki= +/- xx.xxx]	AL4	[CFFKI]
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This function is active with POS. 1.3 see page 27 set on insertion. See manual of insertion sensor for more details

(POS. 1.13) <i>Kp</i>	[Kp = +/- xxxxx]	AL4	[SIDKP]
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This function is active with POS. 1.3 see page 27 set on insertion. See manual of insertion sensor for more details

(POS. 1.14) <i>KC</i>	[KC= +/- xx.xxx]	AL4	[CFFKC]
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Calibration Factor. This function is activated if the sensor model is NOT present on the sensors table standard parameters

(POS. 1.15) <i>Coils Ex.Current</i>	[C.Curr.= mA xxx.x]	AL4	[CEXCC]
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Excitation coils current. This function is activated if the sensor model is NOT present on the sensors table standard parameters

(POS. 1.16) <i>C. Reg. PB</i>	[C.Reg.PB= xxx]	AL4	[CRPRB]
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Current regulator parameter. This function is activated if the sensor model is NOT present on the sensors table standard parameters

(POS. 1.17) <i>C. Reg. DK</i>	[C.Reg. DK = xxx]	AL4	[CRDER]
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Current regulator parameter. This function is activated if the sensor model is NOT present on the sensors table standard parameters

(POS. 1.18) <i>Preamplifier</i>	[Preamplif.= ON]	AL4	[PREIN]
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This function enable/disable the presence of the preamplifier. The use of the preamplifier inhibit the possibility to use BIV and Empty Pipe Detection.

(POS. 1.19) <i>S. Freq.</i>	[S.Freq.= Hz xx]	AL4	[SFREQ]
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Measure sampling frequency. This function is activated if the sensor model is NOT present on the sensors table standard parameters.

(POS. 1.20) <i>Empty Pipe Detection</i>	[E.P.Detect= ON]	AL4	[EPDEN]
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Enables the empty pipe detection function. This function is useful to keep the meter lock to zero when the pipe become empty.

(POS. 1.21) <i>Empty pipe thr.</i>	[R max= Kohm xxxx]	AL4	[EPDTH]
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Maximum resistance value at the inputs (electrodes) determine the empty pipe condition. This feature is enabled only if the "1.19" page 27 Empty Pipe Detection is ON.

(POS. 1.22) <i>Electrodes cleaning</i>	[El. cleaning=AVG]	AL4	[ELCLN]
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This function is useful to keep the electrodes clean; the allowed values are: OFF, minimum, average and maximum. It is **not recommended** to use this function when the liquid has a conductivity less than 100µS/cm (**set to OFF**).

(POS. 1.23) <i>Sensor Cable</i>	[S.cable=m xxx]	AL4	[SCALN]
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Length of cable between sensor and converter on separate version.

(POS. 1.24) <i>Signal error delay</i>	[S.err.delay=m xxx]	AL4	[SEALT]
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Delay before generating error. This function is useful to prevent unexpected lock to zero of measure caused by sporadic events (empty pipe, excitation error, signal error)

(POS. 1.25) <i>Sensor verify</i>	[Sens. verify= OFF]	AL3	[ASVFE]
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Enable the Automatic sensor verification (see BIV optional function).

(POS. 1.26) Zero point calibration [Zero point cal.]

This feature appears only when the process conditions are as follow:

- measure filters recommended (Damping) set to SMART 2 second / 5 second
- stable flow rate and lower than 0.1% of the absolute scale (10 m/s)
- It must have elapsed at least 10 minutes after the last significant change of flow rate

When the above conditions are met, the zero point calibration function will appears on the display, press the "Enter / ESC" and automatically the procedure will start.

NOTICE: Be sure that the the sensor is completely full of liquid and perfectly still. Even subtle movements of the fluid can cause significant errors, therefore proceed with great care.

(POS. 1.27) Coefficient KL [KL=XX +/- XXXXXXXXXX] AL4 [SETKL]

Linearization coefficient for negative flow, reserved to the service. This command is only showed if SMODL = 000.

MENU 1 - SENSOR: ONLY MCP FUNCTIONS**Sensor Coils Time A [MCP ONLY] AL4 [SCTMA]**

Reference sensor coil time A

Sensor Coils Time B [MCP ONLY] AL4 [SCTMB]

Reference sensor coil time B

Sensor Coils Resistance [MCP ONLY] AL4 [SCRES]

Reference sensor coil resistance

Resistance E1 [MCP ONLY] AL4 [SE1RR]

Resistance value of E1 electrodes

Resistance E2 [MCP ONLY] AL4 [SE2RR]

Resistance value E2 electrodes

Sensor Coils Temperature Reference [MCP ONLY] AL4 [SCTRF]

sensor data reference temperature. Temperature measured on the coils of the sensor at the time of saving the instrument reference data for B.I.V. function The temperature value in degrees Celsius, which owns the sensor.

Note: the temperature must be estimated on the basis of the place of the sensor installation.

MENU 2 - UNITS

WARNING: The totalizer value is updated and changed depending on the setting of unit value. The scale change may cause accuracy loss depending of rounding up. For example, if T +=0,234 liters with 3 decimals, it become T +=0.001 m³ losing 0.234 liters in rounding up.

(POS. 2.1) <i>Diameter</i>	[Diam.= mm]	AL2	[SDIUM]
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Sensor diameter unit of measure (mm or inch)

(POS. 2.2) <i>S.cable length unit of m. type</i>	[S.Cable= m]	AL2	[SCAUM]
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Sensor cable length for separate version. Select m or foot.

(POS. 2.3) <i>Flow rate unit of m. type</i>	[FR unit= METRIC]	AL2	[FRMUT]
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Flow rate type measure unit. Select metric or not metric (Imperial units)

(POS. 2.4) <i>Pulse 1 unit of m. type</i>	[PL1 unit= METRIC]	AL2	[PL1UT]
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This function is active with POS. 7.1 see page 29 enable.

This function changes the choice of measure unit POS. 3.4 see page 28

Pulse 1 type measure unit: metric or not metric (Imperial units).

(POS. 2.5) <i>Pulse 2 unit of m. type</i>	[PL2 unit= METRIC]	AL2	[PL2UT]
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This function is active with POS. 7.2 see page 29 enable.

This function changes the choice of measure unit POS. 3.6 see page 28

Pulse 2 type measure unit: metric or not metric Imperial units).

(POS. 2.6) <i>Totalizer direct unit of m. type</i>	[T+ unit= METRIC]	AL2	[TTPUT]
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Setting total direct totalizer measure unit type: metric or not metric Imperial units).

This function changes the values measure unit on POS. 2.7 see page 27

(POS. 2.7) <i>Totalizer direct unit of measure</i>	[T+ unit= dm³]	AL2	[TTPUM]
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Setting total direct totalizer measure unit.

This function visualized on visualization pages.

(POS. 2.8) <i>Totalizer direct Deci.Point pos.</i>	[T+ D.P.= x]	AL2	[TTPDP]
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Setting total direct totalizer decimal point position.

Example: T+D.P.= 3 visualized value T+dm³ 0.000 / T+D.P.= 2 visualized value T+dm³ 0.00

(POS. 2.9) <i>Total. Part. direct unit of m. type</i>	[P+ unit= METRIC]	AL2	[TPPUT]
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This function is active with POS. 9.7 see page 29 enable.

Setting partial direct totalizer measure unit type: metric or not metric (Imperial units).

This function changes the values measure unit on POS. 2.10 see page 27 .

It is visualized on visualization pages.

(POS. 2.10) <i>Total. Partial+ unit of measure</i>	[P+ unit= dm³]	AL2	[TPPUM]
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Setting partial direct totalizer measure unit.

This function visualized on visualization pages.

(POS. 2.11) Total. Part direct Dec. Point pos.	[P+ D.P.= x]	AL2	[TPPDP]
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Setting partial direct totalizer decimal point position.

Example: P+D.P.= 3 visualized value P+dm³ 0.000 / P+D.P.= 2 visualized value P+dm³ 0.00

(POS. 2.12) Total. T reverse unit of m. type	[T- unit= METRIC]	AL2	[TTNUT]
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This function is active with POS. 9.8 see page 29, enabled.

Setting total reverse totalizer measure unit type: metric or not metric (Imperial units).

This function changes the values measure unit on POS. 2.13 see page 27.

It is visualized on visualization pages.

(POS. 2.13) Total. T reverse unit of meas.	[T- unit= dm³]	AL2	[TTNUM]
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Setting total reverse totalizer measure unit.

This function visualized on visualization pages.

(POS. 2.14) Total. T reverse Dec. Point Pos.	[T- D.P.= x]	AL2	[TTNDP]
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Setting total reverse totalizer decimal point position.

Example: T- D.P.= 3 visualized value T- dm³ 0.000

T- D.P.= 2 visualized value T- dm³ 0.00

(POS. 2.15) Total. Part reverse unit of m. type	[P- unit= METRIC]	AL2	[TPNUT]
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This function is active with POS. 9.8 see page 29, enable.

Setting partial reverse totalizer measure unit type: metric or not metric (Imperial units).

This function changes the values measure unit on POS. 2.16 see page 27

It is visualized on visualization pages.

(POS. 2.16) Total. Part reverse unit of measure	[P- unit= dm³]	AL2	[TPNUM]
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Setting partial reverse totalizer measure unit.

This function visualized on visualization pages.

(POS. 2.17) Tot Part. reverse Dec. Point Pos.	[P- D.P.= x]	AL2	[TPNDP]
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Setting partial reverse totalizer decimal point position.

Example: P- D.P.= 3 visualized value P-dm³ 0.000

P- D.P.= 2 visualized value P-dm³ 0.00

(POS. 2.18) Temperature unit of measure	[Temp. unit= C°]	AL2	[TMPUT]
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Setting temperature measure unit.

(POS. 2.19) Mass units enable	[Mass units= ON/OFF]	AL2	[MSSUE]
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Enable or Disable the selection of mass unit of full scale set.

(POS. 2.20) Specific Gravity coeffic.	[Sg= Kg/dm³ x.xxxx]	AL2	[VMSGC]
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Setting specific gravity coefficient. This Function is active with POS. 2.19 see page 27, enable.